

PREIMAGE-CHAIN CORRECTIONS, PRIMITIVE-BLOCK RIGIDITY, AND NEW LOWER BOUNDS FOR SUPERPERMUTATIONS

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ABSTRACT. Building on the Lean-verified component method of Hunter and Raudvere, this paper quantifies the additional cost imposed when a spanning system of strongly exitless components is realized as the Hunter image of a Hamilton path in the permutation-overlap graph.

A component-matching theorem for the auxiliary preimage-chain graph identifies its source and target components up to one global exception. At the source end, all Hunter components are represented except for the source/terminal exception. At the target end, all non-root Hunter components admit a capacity-one matching, augmented by a terminal symbol when required. This structure yields an exact *preimage-chain identity* that decomposes $\text{wt}(P) - B_1(F(P))$ into nonnegative root, terminal, chain, and cycle terms. A shortest-path and assignment relaxation defines a computable functional Π satisfying

$$\text{wt}(P) \geq B_1(F(P)) + \Pi(F(P)).$$

For $k = 6$, an explicit Hunter image has $B_1 = 863$ and $\Pi = 6$, whereas a known 872-letter construction has $B_1 = 866$ and $\Pi = 0$.

A σ^2 -reduced normal form, a rate-defect normal form, and an endpoint-matching capacity theorem are obtained for exact weight-three interval-piece chains. They imply the primitive-block stability inequality

$$(k-2)t \leq (k-2)D_k\pi + (D_k-1)d, \quad D_k = k^2 - 3k + 1,$$

where $\pi = 1$ precisely when a component has minimum entry weight two and terminates in a full interval piece. In a reduced preimage, the associated terminal portal cannot be absorbed at zero cost by any minimum-entry-two target, including a self-nested target.

The pointwise actual-image Hunter bound assigns a canonical nonnegative integer layer to every Hamilton path. Portal rigidity, defect-budget monotonicity, the root-biased distinguished-component envelope, and capacity-one target matching then give

$$S(k) \geq \mathcal{H}_k + \Gamma_k, \quad \Gamma_k = \left(1 - \frac{3}{k} + O(k^{-2})\right) (k-4)!.$$

In particular,

$$S(7) \geq 5892, \quad S(8) \geq 46118, \quad S(14) \geq 93,890,256,441.$$

The arbitrary- k proof chain, including actual component capacity, the pointwise actual-image baseline, the portal assignment, σ^2 normalization, the unconditional pathwise theorem, the global $S(k)$ theorem, and the numerical corollaries for $k = 5, \dots, 14$, has been formalized in Lean 4.31.0 against a pinned Hunter–Raudvere library. The formal development contains no unfinished proof or project-local axiom. A separate deterministic Python implementation reproduces the finite certificates, local transcriptions, and exact integer tables.

An explicit two-block bridge family further shows that every single-component, single-portal linear envelope of the form $t(C) \leq D_k\pi(C) + c_k d(C)$ must satisfy $c_k = \Omega(k)$ for $k \geq 8$. This obstruction does not apply to nonlinear, multistate, prefix-sensitive, intercomponent, or iterated-transform methods.

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1. INTRODUCTION

A *superpermutation* on k symbols is a finite word over a k -letter alphabet in which every permutation of the alphabet occurs as a contiguous factor. Let $S(k)$ denote the minimum possible length. The values $S(1), \dots, S(5)$ are known, whereas the problem remains open for every $k \geq 6$. Ashlock and Tillotson gave the classical recursive construction of length $\sum_{i=1}^k i!$ [AT93]; Johnston established non-uniqueness at the formerly conjectured length [Joh13]; and Houston constructed a six-symbol superpermutation of length 872 [Hou14]. Egan’s construction, based on Williams’s Hamilton path in a two-generator Cayley digraph [Wil17, Ega18], yields

$$(1) \quad S(k) \leq k! + (k-1)! + (k-2)! + (k-3)! + k - 3.$$

A formally verified gain-one lift lowers this upper bound by one for every $k \geq 8$ [Rau26].

The present work builds on the lower-bound framework of Hunter and Raudvere [HR26], whose complete theorem chain has been formalized in Lean. Define

$$(2) \quad D_k = k^2 - 3k + 1, \quad A_k = k^3 - 2k^2 - k - 1.$$

Their exact pointwise lower bound for superpermutation length is

$$(3) \quad \mathcal{H}_k = k! + (k-1)! + (k-2)! + k - 3 + \left\lceil \frac{(k-2)! - (k-2)}{D_k} \right\rceil.$$

In particular, $\mathcal{H}_6 = 869$, $\mathcal{H}_7 = 5888$, and $\mathcal{H}_8 = 46103$. The same formal development characterizes the image of the Hunter transformation as the class of spanning unions of strongly exitless paths; this characterization is essential in the six-symbol analysis below.

Hunter's bound prices the transformed components separately. The central question is therefore not whether the estimate for an isolated strongly exitless component can be sharpened, but how much additional cost is imposed by the simultaneous realization of several such components as the image of a single Hamilton path. The analysis consists of the following components.

- (1) A preimage-chain component-matching theorem, an exact identity for $\text{wt}(P) - B_1(F(P))$, and a computable shortest-path/assignment relaxation II.
- (2) A global σ^2 -reduced normal form that replaces every zero-cost decomposable double rotation by a proper door without increasing path weight.
- (3) The rate-defect normal form

$$d = D_k(x + \mu - 2) + (k-2)\Delta - r, \quad t = D_k m - (k-1)d.$$

- (4) An endpoint-matching capacity theorem and the primitive-block stability inequality, including the self-nested terminal-portal case.
- (5) A pointwise actual-image Hunter baseline that assigns every Hamilton path a canonical nonnegative integer layer $s(P)$.
- (6) An explicit layerwise correction derived from terminal-portal rigidity, with one global exception and any distinguished non-root target charged to the exact root gap.
- (7) Equality of the actual Hunter image class and a locally defined relaxed component class, established through the formalized image characterization.
- (8) Two limitations of the method: for $k \geq 8$, a two-block bridge family forces $c_k = \Omega(k)$ in every envelope $t(C) \leq D_k \pi(C) + c_k d(C)$, while neutral weight-three refinement creates no independent block-level obligation.

The main numerical consequences are summarized below.

Theorem 1.1 (Lean-formalized numerical corollaries). *The unconditional formal theorem gives*

$$\begin{array}{ll} S(5) \geq 153, & S(6) \geq 869, \\ S(7) \geq 5,892, & S(8) \geq 46,118, \\ S(9) \geq 408,418, & S(10) \geq 4,033,080, \\ S(11) \geq 43,916,235, & S(12) \geq 522,610,764, \\ S(13) \geq 6,746,523,219, & S(14) \geq 93,890,256,441. \end{array}$$

The values for $k = 7, \dots, 14$ are the new lower bounds of this paper; $k = 5, 6$ are included to state the complete formally exported table.

The arbitrary- k structural results in the main proof chain are formalized in Lean 4.31.0. The formal development is published in the

[public GitHub repository](#).

The exact formalization source snapshot is commit

a3852f0de2d84b99cc65d4c281e86892b5f6e911.

A separate deterministic Python implementation reproduces the finite transcriptions and certificate checks; its role is computational audit rather than a second theorem-prover formalization or external peer review.

2. THE OVERLAP GRAPH AND HUNTER'S COMPONENT CLASSES

Let $\mathfrak{S}_k$ be the symmetric group written as words of length k . For $u, v \in \mathfrak{S}_k$, define

$$(4) \quad w(u, v) = \min\{d \in \{1, \dots, k\} : u_{d+1} \cdots u_k = v_1 \cdots v_{k-d}\}.$$

For a Hamilton path $P = (p_1, \dots, p_k)$, put

$$\text{wt}(P) = \sum_{i=1}^{k!-1} w(p_i, p_{i+1}).$$

The corresponding word has length $k + \text{wt}(P)$, and hence

$$(5) \quad S(k) = L(k) + k,$$

where $L(k)$ is the minimum Hamilton-path weight.

Write a word as $u = abR$, with $|R| = k - 2$. Define

$$(6) \quad \sigma(u) = bRa, \quad \sigma^2(u) = Rab, \quad \tau(u) = Rba.$$

Thus σ is the weight-one rotation and τ is the proper weight-two door. The σ -orbits are the *rotation classes*; each has k vertices, and there are $(k-1)!$ classes.

An *early exit* occurs when a path leaves a rotation class along an edge of weight at least two before visiting the weight-one successor of the current vertex. A path with no early exit is *exitless*; it is *strongly exitless* if, in addition, the weight-one successor of its final vertex already lies on the path.

Let $\mathcal{X}_k$ denote the actual image class

$$\mathcal{X}_k = \{F(P) : P \text{ is a Hamilton path}\}.$$

Let $\mathcal{Y}_k$ denote the class of spanning directed graphs whose weak components are vertex-simple strongly exitless paths. The following statement combines the necessity and sufficiency directions formalized by Hunter and Raudvere.

Theorem 2.1 (Hunter–Raudvere image characterization and component bound). *For $k \geq 2$,*

$$\boxed{\mathcal{X}_k = \mathcal{Y}_k.}$$

For every Hamilton path P and $X = F(P)$:

- (i) *every component of X is a union of complete rotation classes;*
- (ii) *the tail t_C of a component C is the first vertex of C visited by P ;*
- (iii) *with $\mu_C = \text{minto}(C)$,*

$$(7) \quad \text{wt}(P) \geq B_1(X) := |\text{Comp}(X)| - 1 + \text{wt}(X) + \sum_C \mu_C - \max_C \mu_C;$$

- (iv) *if a non-spanning component contains t rotation classes, then*

$$(8) \quad \text{pValue}(C) := 1 + \mu_C + \text{wt}(C) \geq \frac{A_k}{D_k} t.$$

Moreover, their pointwise component theorem gives the corresponding lower bound on $B_1(X)$ for every $X \in \mathcal{X}_k$.

For every component write the signed rate deviation

$$\delta(C) = D_k \text{pValue}(C) - A_k t(C) \in \mathbb{Z}.$$

For a non-spanning component, (8) implies $\delta(C) \geq 0$, and the *rate defect* is defined by

$$(9) \quad \boxed{d(C) = D_k \text{pValue}(C) - A_k t(C) \geq 0.}$$

No nonnegative defect is assigned to a spanning component. Under the standard empty-exterior convention $\mu(C) = 0$, the signed deviation of a spanning component can be negative; for example, the classical 153-letter order-five path has $\delta(C) = -17$.

A component is always a vertex-simple path. A *self-nested* preimage chain is one whose source and target are the same component; it is *self-closing* if the forced target vertex equals the component tail. A *terminal portal* will mean a component with minimum entry two and a full final interval piece. Primitive ladders and bridge blocks are defined in [Section 11](#).

3. A σ^2 -REDUCED NORMAL FORM

The two weight-two successors in (6) play different roles in the portal analysis. The required normal form is stated below.

Lemma 3.1 (Classification of weight-two successors). *Let $k \geq 3$ and $u, v \in \mathfrak{S}_k$. Then $w(u, v) = 2$ if and only if*

$$v = \sigma^2 u \quad \text{or} \quad v = \tau u.$$

Proof. Write $u = abR$. An overlap of length $k - 2$ forces v to begin with R . Its final two letters must be the two missing symbols a, b , in one of the two orders. These give $Rab = \sigma^2 u$ and $Rba = \tau u$. Both have weight exactly two because the symbols of u are distinct. $\square$

Definition 3.2. *A Hamilton path is σ^2 -reduced if it contains no edge $u \rightarrow \sigma^2 u$.*

For a path P , let $N_\sigma(P)$ be the number of its weight-one rotation edges $u \rightarrow \sigma u$.

Lemma 3.3 (σ^2 surgery). *Let $k \geq 3$. If a Hamilton path P contains $u \rightarrow \sigma^2 u$, then there is a Hamilton path P' such that*

$$\text{wt}(P') \leq \text{wt}(P), \quad N_\sigma(P') \geq N_\sigma(P) + 2.$$

Proof. Put $v = \sigma u$ and $w = \sigma^2 u$. The vertex v occurs somewhere in P . Remove it from its old position, and then insert it between the consecutive vertices u, w . Removing a vertex cannot increase path weight, by the triangle inequality for overlap weights. The insertion preserves the weight of the edge $u \rightarrow w$, because

$$w(u, w) = 2 = w(u, v) + w(v, w) = 1 + 1.$$

The resulting list is a permutation of the original Hamilton list and is therefore again a Hamilton path.

It remains to prove strict progress. At its old position, v cannot have predecessor u : the path already uses the successor w of u . Likewise, v cannot have successor w : the path already uses the predecessor u of w . Since u is the unique vertex whose rotation successor is v , and w is the unique rotation successor of v , neither old edge incident with v is a rotation edge. Deleting v therefore destroys no edge counted by N_σ , while inserting it creates the two rotation edges $u \rightarrow v$ and $v \rightarrow w$. The splice at the old location may create another rotation edge, but cannot reduce the stated gain. $\square$

Theorem 3.4 (σ^2 -reduced normal form). *For every Hamilton path P with $k \geq 3$, there exists a σ^2 -reduced Hamilton path P^* with*

$$\text{wt}(P^*) \leq \text{wt}(P).$$

Consequently, a minimum-weight Hamilton path may be chosen σ^2 -reduced.

Proof. Repeatedly apply [Lemma 3.3](#) while a σ^2 edge remains. At each step N_σ increases by at least two, while always

$$0 \leq N_\sigma(P) \leq k! - 1.$$

The process therefore terminates. Weight never increases. If P was minimum-weight, the final path has the same minimum weight. $\square$

4. THE PREIMAGE-CHAIN IDENTITY

Let P be a Hamilton path and $X = F(P)$. Hunter's construction identifies a set $S(P)$ of vertices. For every $v \in S(P)$, it removes the path edge leaving v and inserts $v \rightarrow \sigma v$; if this creates indegree two, it removes the original path edge entering the doubled vertex. Let E_1 and E_2 be the first and second sets of removed path edges, and put

$$I = \mathbf{1}[p_{k!} \in S(P)].$$

Hunter's exact bookkeeping is

$$(10) \quad \text{wt}(P) = \text{wt}(X) + \sum_{e \in E_1} (w(e) - 1) - I + \sum_{e \in E_2} w(e).$$

Define a directed graph D_P on $S(P)$ by

$$(11) \quad u \longrightarrow_{D_P} v \iff (u, \sigma v) \text{ is an edge of } P.$$

Every vertex of D_P has indegree and outdegree at most one, so D_P is a disjoint union of directed paths and cycles. Write $C_{-1}(P)$ for the rotation predecessors of the first-visited vertices of the rotation classes. The defining exclusion in Hunter's construction is

$$S(P) \cap C_{-1}(P) = \emptyset.$$

Let $\mathcal{K}(P)$ be the set of directed path components of D_P . For $K = (v_1, \dots, v_r) \in \mathcal{K}(P)$, call K *terminal* when $v_r = p_{k!}$. When K is nonterminal, its source and target Hunter components are denoted by $B(K)$ and $A(K)$, respectively.

Theorem 4.1 (Preimage-chain component matching). *Let P be a Hamilton path, $X = F(P)$, and let C_0 be the component of X containing p_1 .*

- (i) *Every $K = (v_1, \dots, v_r) \in \mathcal{K}(P)$ has a unique initial edge*

$$h_{B(K)} \longrightarrow \sigma v_1 \in E_2,$$

where $h_{B(K)}$ is the head of a unique component $B(K)$ of X .

- (ii) *Every nonterminal K has a unique exit edge*

$$v_r \longrightarrow t_{A(K)} \in E_1,$$

where $t_{A(K)}$ is the tail of a unique non-root component $A(K)$ of X .

- (iii) *If $I = 0$, there is no terminal chain. Let L be the component containing $p_{k!}$. Then*

$$B : \mathcal{K}(P) \xrightarrow{\sim} \text{Comp}(X) \setminus \{L\}, \quad A : \mathcal{K}(P) \xrightarrow{\sim} \text{Comp}(X) \setminus \{C_0\}$$

are bijections.

(iv) If $I = 1$, there is exactly one terminal chain K_* . The source map is a bijection

$$B : \mathcal{K}(P) \xrightarrow{\sim} \text{Comp}(X),$$

and, after setting $\hat{A}(K_*) = \star$ and $\hat{A}(K) = A(K)$ otherwise,

$$\hat{A} : \mathcal{K}(P) \xrightarrow{\sim} (\text{Comp}(X) \setminus \{C_0\}) \cup \{\star\}$$

is a bijection.

In particular, every Hunter component is the source of at most one preimage chain, and every non-root component is the target of at most one nonterminal preimage chain. This remains true for self-nested chains, for which $A(K) = B(K)$.

Proof. Let $K = (v_1, \dots, v_r)$ be a path component of D_P . Since $v_1 \in S(P)$ and $S(P)$ excludes $C_{-1}(P)$, the vertex σv_1 cannot be the first vertex p_1 . It therefore has a unique predecessor b in P . The absence of a D_P -predecessor of v_1 implies $b \notin S(P)$. Hence the original edge $b \rightarrow \sigma v_1$ survives the first deletion step, while the added edge $v_1 \rightarrow \sigma v_1$ gives σv_1 indegree two. The former edge is therefore removed in E_2 . Since $b \notin S(P)$, no added edge leaves b , and b is the head of a unique component of X . This proves the existence and uniqueness in (i).

Conversely, let h be a head of X . If $h \neq p_{k!}$, then h has a unique successor in P . It cannot lie in $S(P)$, because an added rotation edge would then leave it. Its original path edge is not removed in E_1 and must therefore be removed in E_2 . Thus it has the form $h \rightarrow \sigma v$ for a unique $v \in S(P)$. If v had a D_P -predecessor $u \in S(P)$, then the uniqueness of the predecessor of σv in P would force $u = h$, contradicting $h \notin S(P)$. Hence v is a D_P -path start. The path endpoint $p_{k!}$ is a head of X exactly when $p_{k!} \notin S(P)$, that is, exactly when $I = 0$. This proves the source bijections in (iii) and (iv).

Suppose that K is nonterminal, and let a be the successor of v_r in P . Since $v_r \in S(P)$, the edge $v_r \rightarrow a$ belongs to E_1 . If $a = \sigma u$ for some $u \in S(P)$, then $v_r \rightarrow_{D_P} u$, contrary to the maximality of K . Thus no added rotation edge enters a , and after the unique original incoming edge is removed, a is a tail of X . It is not p_1 , so it is the tail of a non-root component. This proves (ii).

Conversely, let $a \neq p_1$ be a tail of X , and let v be its unique predecessor in P . The edge $v \rightarrow a$ cannot have been removed only in E_2 , because an E_2 removal occurs precisely when an added rotation edge still enters the target, whereas a has indegree zero in X . Hence $v \rightarrow a \in E_1$, so $v \in S(P)$. Since no added edge enters a , there is no $u \in S(P)$ with $a = \sigma u$; consequently v is the endpoint of a unique D_P path. The first vertex p_1 is always an X -tail: it has no original predecessor, and an added edge $v \rightarrow p_1$ would imply $v \in C_{-1}(P) \cap S(P)$. Thus the nonterminal target map is a bijection onto $\text{Comp}(X) \setminus \{C_0\}$.

Finally, $p_{k!}$ is a vertex of D_P exactly when $I = 1$. In that case it has no D_P -successor and lies in a unique terminal path component; when $I = 0$ no terminal path exists. Directed cycles of D_P have neither starts nor ends and correspond only to internal E_1 edges, so they do not affect either matching. The stated bijections follow. $\square$

If $c = |\text{Comp}(X)|$, the theorem gives exactly $c - 1$ nonterminal chains and $c - 1 + I$ chains in total. Directed cycles have no component-level source or target and contribute only their internal E_1 edges.

For a nonterminal chain define

$$(12) \quad g(K) = [w(h_{B(K)}, \sigma v_1) - 2] + \sum_{i=1}^{r-1} [w(v_i, \sigma v_{i+1}) - 1] \\ + [w(v_r, t_{A(K)}) - \mu_{A(K)}].$$

For a terminal chain omit the last bracket. For a directed cycle $Z = (v_1, \dots, v_r)$, put

$$(13) \quad g(Z) = \sum_{i=1}^r [w(v_i, \sigma v_{i+1}) - 1], \quad v_{r+1} = v_1.$$

Every bracket is nonnegative.

Theorem 4.2 (Exact preimage-chain identity). *Let C_0 be the component of X containing the first vertex of P , and let $\mu_{\max} = \max_C \mu_C$. Then*

$$(14) \quad \boxed{\text{wt}(P) - B_1(X) = \mu_{\max} - \mu_{C_0} + I + \sum_K g(K) + \sum_Z g(Z).}$$

Proof. By Theorem 4.1, the initial edges of the D_P path components are exactly the E_2 boundary edges, there are $c - 1 + I$ of them, and the nonterminal exits are in bijection with the components $A \neq C_0$. Partition E_1 into internal chain edges, those nonterminal exits, and cycle edges. Extract a baseline two from every E_2 edge and a baseline $\mu_A - 1$ from the unique exit into every $A \neq C_0$. Including the $-I$ in (10), the extracted baseline is

$$2(c - 1 + I) + \sum_{A \neq C_0} (\mu_A - 1) - I.$$

It equals

$$(c - 1) + \sum_A \mu_A - \mu_{\max} + (\mu_{\max} - \mu_{C_0}) + I.$$

The first three terms, together with $\text{wt}(X)$, form $B_1(X)$; the remaining edge terms are precisely the quantities in (12) and (13). Directed cycles contribute only the terms $g(Z)$ and no component boundary term. $\square$

4.1. A shortest-path and assignment relaxation. Let

$$\Omega_X = \{v : (v, \sigma v) \in X\}$$

be the set of rotation slots. For a component head B , slots $u \neq v$, and a component tail A , define

$$a_B(v) = w(h_B, \sigma v) - 2, \quad b(u, v) = w(u, \sigma v) - 1, \\ c_A(v) = w(v, t_A) - \mu_A, \quad v \notin A.$$

All are nonnegative integers. Work in the extended nonnegative integers

$$\bar{\mathbb{N}} = \mathbb{N} \cup \{+\infty\}.$$

Let $d_X(B, A)$ be the minimum cost

$$a_B(v_1) + \sum_{i < r} b(v_i, v_{i+1}) + c_A(v_r)$$

over nonempty slot sequences with $v_r \notin A$; if no sequence exists, set $d_X(B, A) = +\infty$. Define $d_X(B, \star)$ similarly, omitting the exit term.

For $I = 0$, choose a candidate root R , a candidate last component L , and a bijection

$$\phi : \text{Comp}(X) \setminus \{L\} \longrightarrow \text{Comp}(X) \setminus \{R\}.$$

Set

$$\Pi_0(X) = \min \left(\mu_{\max} - \mu_R + \sum_{B \neq L} d_X(B, \phi(B)) \right).$$

For $I = 1$, choose R and a bijection

$$\phi : \text{Comp}(X) \longrightarrow (\text{Comp}(X) \setminus \{R\}) \cup \{\star\},$$

and set

$$\Pi_1(X) = 1 + \min \left(\mu_{\max} - \mu_R + \sum_B d_X(B, \phi(B)) \right).$$

Finally, $\Pi(X) = \min(\Pi_0(X), \Pi_1(X))$.

Corollary 4.3 (Preimage-chain strengthening). *For every Hamilton path P ,*

$$\text{wt}(P) \geq B_1(F(P)) + \Pi(F(P)).$$

Thus $B_1 + \Pi$ pointwise dominates B_1 and is strictly larger on explicit images, though equality may hold on record constructions.

Proof. When $I = 0$, [Theorem 4.1](#) gives bijections from the actual chains to $\text{Comp}(X) \setminus \{L\}$ at the source end and to $\text{Comp}(X) \setminus \{C_0\}$ at the target end; composing them gives a feasible assignment in the definition of Π_0 . When $I = 1$, the same theorem gives the feasible assignment to $(\text{Comp}(X) \setminus \{C_0\}) \cup \{\star\}$ required by Π_1 . The actual slot sequences are feasible routes for the corresponding d_X terms. Replacing each route by an individual shortest route, allowing different routes to reuse slots, and deleting nonnegative cycle costs can only lower the right-hand side of (14). $\square$

5. THE SIX-SYMBOL CASE: SHARPNESS AND PREIMAGE CORRECTION

The accompanying certificate consists of eight disjoint strongly exitless paths spanning $\mathfrak{S}_6$, with component sizes

$$4 \times 114, \quad 2 \times 108, \quad 2 \times 24.$$

Their weights and minimum entry weights are

multiplicity	$ C $	$\text{wt}(C)$	μ_C
4	114	134	2
2	108	127	2
2	24	26	2.

Hence

$$B_1(X) = 7 + 842 + (16 - 2) = 863.$$

The certificate verifier checks spanning, disjointness, simplicity, and strong exitlessness, so $X \in \mathcal{Y}_6$. By the formalized sufficiency direction in [Theorem 2.1](#), $X \in \mathcal{X}_6$ and therefore has at least one Hamilton preimage. The Hunter pointwise component theorem gives $B_1(Y) \geq 863$ for every $Y \in \mathcal{X}_6$. Consequently

$$(15) \quad \boxed{\min_{Y \in \mathcal{X}_6} B_1(Y) = 863.}$$

For this fixed image, the assignment relaxation gives

$$\Pi_0(X) = 6, \quad \Pi_1(X) = 7.$$

Thus every Hamilton path P with $F(P) = X$ has weight at least 869, and every word obtained from such a preimage has length at least 875. This is a statement about the fixed image X ; it is *not* the global assertion $S(6) \geq 875$.

By contrast, a known 872-letter word has path weight 866, component-size distribution

$$25 \times 24 + 1 \times 120,$$

and

$$B_1(F(P)) = 866, \quad \Pi(F(P)) = 0.$$

The correction therefore removes an independently priced false extremum while remaining exact on the known record construction.

6. RATE DEFECTS AND INTERVAL-PIECE GEOMETRY

Throughout the remainder of the structural argument, assume $k \geq 5$. The interval-piece formulas below apply to every strongly exitless component. Whenever the nonnegative rate defect $d(C)$ is used, the component is explicitly assumed non-spanning. A strongly exitless component visits each rotation class consecutively and completely. Between successive classes, a weight-two edge inside a reduced component is a proper door; maximal runs connected by such doors are called *interval pieces*. Let

- $t = t(C)$ be the number of rotation classes;
- $r = r(C)$ be the number of interval pieces;
- $\Delta = (k - 1)r - t$ be the total piece deficit;
- x be the sum over piece seams of (seam weight $- 3$);
- $\mu = \text{minto}(C)$.

A piece is *full* if it contains $k - 1$ classes; otherwise it is partial. Every class contributes $k - 1$ rotation edges, the $t - r$ within-piece doors have weight two, and the $r - 1$ seams have total weight $3(r - 1) + x$. Therefore

$$(16) \quad \text{wt}(C) = (k + 1)t + r + x - 3, \quad \text{pValue}(C) = (k + 1)t + r + x + \mu - 2.$$

Proposition 6.1 (Rate-defect normal form). *Let C be a non-spanning strongly exitless component and put $d = d(C)$. Then*

$$(17) \quad \boxed{d = D_k(x + \mu - 2) + (k - 2)\Delta - r.}$$

If

$$(18) \quad m = \Delta + (k - 1)(x + \mu - 2),$$

then

$$(19) \quad \boxed{t = D_k m - (k - 1)d, \quad \text{pValue}(C) = A_k m - k^2 d.}$$

In particular,

$$t \equiv -(k - 1)d \pmod{D_k}.$$

Proof. Substitute (16) into $d = D_k \text{pValue} - A_k t$ and use

$$A_k = (k + 1)D_k + k - 2, \quad D_k = (k - 1)(k - 2) - 1, \quad t = (k - 1)r - \Delta.$$

Solving the resulting linear equations gives (19). $\square$

7. PORTAL-RUN CAPACITY FOR INTERVAL-PIECE CHAINS

An exact weight-three chain is a sequence of interval pieces with pairwise disjoint supports and weight-three seams. A partial piece has deficit $\delta \geq 1$.

Lemma 7.1 (Portal-run local geometry). *In an exact weight-three chain:*

- (i) a pure full run has length at most $k - 2$;
- (ii) a deficit-one partial piece cannot be adjacent to positive full runs on both sides;
- (iii) every endpoint full run adjacent to a partial piece has length at most $k - 3$;
- (iv) an internal full run has length at most $k - 4$, except that length $k - 3$ is possible only when the adjacent deficits a, b satisfy $a + b \geq k - 1$.

The complete arbitrary- k coordinate proof, including all six exact weight-three successor orientations and the cyclic omitted-arc lemma, is given in [Appendix A](#). The finite program serves only as an independent transcription audit of this arbitrary- k proof.

Let a chain contain n_1 deficit-one partial pieces and n_2 partial pieces of deficit at least two. Let $e \in \{0, 1, 2\}$ be the number of chain endpoints occupied by full pieces, R the number of positive full runs, F the number of full pieces, and N the number of internal runs of length $k - 3$.

Theorem 7.2 (Endpoint-matching capacity). *For every exact weight-three chain with r pieces and total deficit Δ ,*

$$(20) \quad 2r \leq (k - 2)(\Delta + e).$$

Proof. The one-sidedness of deficit-one portals gives

$$2R - e \leq n_1 + 2n_2 \leq \Delta,$$

so $R \leq (\Delta + e)/2$. Every full run has baseline length at most $k - 4$; endpoint runs and the N long internal runs may contribute one extra full piece. Hence

$$F \leq (k - 4)R + e + N.$$

Write $p = n_1 + n_2$ and $E = \Delta - p = \sum(\delta_i - 1)$. Each long internal run consumes at least $k - 3$ units of adjacent excess, and each unit is counted by at most two neighboring runs, so $(k - 3)N \leq 2E$. Thus

$$r = p + F \leq \Delta - E + \frac{k - 4}{2}(\Delta + e) + e + \frac{2E}{k - 3}.$$

Subtracting from $(k - 2)(\Delta + e)/2$ leaves $E(k - 5)/(k - 3) \geq 0$. The pure full case has $\Delta = 0$, $e = 2$, and $r \leq k - 2$ by Lemma 7.1. $\square$

If a component has seam residual x , cut the positive-residual seams. If $a, b \in \{0, 1\}$ indicate whether its first and last pieces are full, the total number of full endpoints over the resulting chains is at most $2x + a + b$. Summing (20) yields

$$(21) \quad 2r \leq (k - 2)(\Delta + 2x + a + b).$$

8. PRIMITIVE-BLOCK STABILITY AND RIGID TERMINAL PORTALS

Definition 8.1 (Terminal primitive-block certificate). *For a component C , define*

$$(22) \quad \pi(C) = \begin{cases} 1, & \mu(C) = 2 \text{ and the last interval piece of } C \text{ is full,} \\ 0, & \text{otherwise.} \end{cases}$$

When $\pi(C) = 1$, the final full piece certifies one affine capacity unit of size D_k . This certificate is termed a terminal primitive block; it is attached to the component and does not represent an additional weak component produced by cutting the path.

Theorem 8.2 (Full primitive-block stability). *For every non-spanning strongly exitless component C ,*

$$(23) \quad 2d(C) \geq (k - 2)(m(C) - \pi(C)).$$

Equivalently,

$$(24) \quad (k - 2)t(C) \leq (k - 2)D_k\pi(C) + (D_k - 1)d(C).$$

Proof. From (17), (18), and (21),

$$\begin{aligned} 2d - (k - 2)m &= (D_k - 1)(x + \mu - 2) + (k - 2)\Delta - 2r \\ &\geq (k - 1)(k - 4)x + (D_k - 1)(\mu - 2) - (k - 2)(a + b). \end{aligned}$$

If $\mu = 2$, the first piece is partial: if it were full, both weight-two predecessors of its tail would lie in that first full structure, contradicting $\mu = 2$. Hence $a = 0$ and $b = \pi$, so the right-hand side is

at least $-(k-2)\pi$. If $\mu \geq 3$, then $\pi = 0$ and $a + b \leq 2$; the worst case $x = 0$, $\mu = 3$, $a + b = 2$ leaves

$$(D_k - 1) - 2(k - 2) = (k - 1)(k - 4) \geq 0.$$

This proves (23); substituting $t = D_k m - (k - 1)d$ gives (24). $\square$

8.1. Reduced zero-cost chains. Let P be σ^2 -reduced and let C be a component of $F(P)$ with head h_C . A zero-cost preimage chain has one slot, because every internal transition between distinct slots costs at least one.

Lemma 8.3 (Reduced zero-cost dichotomy). *Suppose a zero-cost single-slot chain leaves C and enters a component A with $\mu_A = 2$. Then its slot and target tail are forced to be*

$$(25) \quad q = \sigma^{-1}\tau(h_C), \quad t_A = \Psi(h_C) := \tau(\sigma^{-1}\tau(h_C)).$$

If the rotation class of $\Psi(h_C)$ lies in C , then either $A = C$ and $t_C = \Psi(h_C)$, or no such zero-cost chain exists.

Proof. Zero entry cost gives $w(h_C, \sigma q) = 2$. By Lemma 3.1, σq is either $\sigma^2 h_C$ or τh_C . The first alternative is an edge of the reduced path P and is excluded; hence $q = \sigma^{-1}\tau h_C$. Zero exit cost gives $w(q, t_A) = 2$. The same classification and reducedness exclude $t_A = \sigma^2 q$, leaving $t_A = \tau q = \Psi(h_C)$.

If $A \neq C$, then A contains the complete rotation class of its tail, which is already contained in C ; this contradicts disjointness of the components. If $A = C$, equality of rotation classes is insufficient: the actual target vertex must equal the component tail, namely $t_C = \Psi(h_C)$. $\square$

Lemma 8.4 (Terminal-full rigidity). *A simple component whose last piece is full cannot satisfy $t_C = \Psi(h_C)$.*

Proof. Write the start of the last full piece as Cm , where m is its marker and $|C| = k - 1$. Its head is $m\sigma^{-1}(C)$. Direct substitution in (25) gives

$$\Psi(h_C) = \sigma(C)m.$$

The last full piece contains the rotation class $[m\sigma(C)]$ and visits every vertex of that class, including $\sigma(C)m$. If this vertex were also the tail of C , the path would repeat a vertex. $\square$

Corollary 8.5 (Rigid terminal portal). *In a σ^2 -reduced Hamilton path, a component counted by $\pi(C)$ cannot enter any minimum-entry-weight-two component at zero preimage-chain cost, including itself. Its only zero-cost exceptions are being the final component or entering a component with minimum entry at least three.*

9. LAYERWISE LOWER BOUNDS

Let

$$(26) \quad B_H(k) = \mathcal{H}_k - k$$

be the Hunter path lower bound, and define

$$(27) \quad \kappa_k = D_k \left\lceil \frac{(k-2)! - (k-2)}{D_k} \right\rceil - ((k-2)! - (k-2)), \quad 0 \leq \kappa_k < D_k.$$

Lemma 9.1 (Pointwise actual-image baseline). *For every $k \geq 5$ and every Hamilton path P ,*

$$(28) \quad B_1(F(P)) \geq B_H(k).$$

Consequently

$$(29) \quad s(P) := B_1(F(P)) - B_H(k) \in \mathbb{N}, \quad B_1(F(P)) = B_H(k) + s(P).$$

Proof. The concrete image $F(P)$ belongs to the actual image class $\mathcal{X}_k$, so the pointwise Hunter–Raudvere reduction in [Theorem 2.1](#) applies to this particular image. Using the pivot strongly exitless path of length $k(k-1)(k-2)$ gives

$$B_1(F(P)) \geq k! + (k-1)! + (k-2)! - 3 + \frac{(k-2)! - (k-2)}{D_k}.$$

The left-hand side is an integer. Taking the ceiling of the final quotient yields (28), because the resulting right-hand side is exactly $\mathcal{H}_k - k$. Equation (29) follows by subtraction in $\mathbb{N}$. $\square$

For the rest of this section fix a Hamilton path P , put $X = F(P)$ and $s = s(P)$, and set

$$(30) \quad K_s = \kappa_k + sD_k.$$

Let C_0 be the actual root component containing the first path vertex. Choose a maximum- μ component C^* as *distinguished*, choosing C_0 when it is tied for the maximum. If X has one component, set $K = 0$. Otherwise every non-distinguished component is non-spanning, and K denotes the sum of their nonnegative rate defects. Write the parameters of C^* as t_*, r_*, Δ_*, x_* . Substitution in the layer equation gives

$$(31) \quad K_s - K = D_k(r_* + x_*) - (k-2)(t_* - 1).$$

Lemma 9.2 (Defect-budget monotonicity). *For every layer decomposition,*

$$(32) \quad 0 \leq K \leq K_s.$$

More precisely,

$$(33) \quad K_s - K \geq (k-1)(k-3)x_* + \frac{k-2}{2}\Delta_*.$$

Proof. Using $t_* = (k-1)r_* - \Delta_*$, rewrite (31) as

$$K_s - K = D_k x_* + (k-2)\Delta_* - r_* + (k-2).$$

By (21) and $a + b \leq 2$,

$$r_* \leq \frac{k-2}{2}\Delta_* + (k-2)x_* + (k-2).$$

Substitution gives (33). Since K is a sum of nonnegative defects, (32) follows. $\square$

Theorem 9.3 (Distinguished-component envelope). *For every layer $s \geq 0$,*

$$(34) \quad t_* \leq (k-1)(k-2) + (k-1)(K_s - K),$$

and

$$(35) \quad (k-2)t_* + (D_k - 1)K \leq C_s,$$

where

$$(36) \quad C_s = (k-1)(k-2)^2 + (D_k + 1)K_s.$$

Proof. The component-capacity bound gives

$$t_* = (k-1)r_* - \Delta_* \leq (k-1)(k-2) + (k-1)(k-2)x_* + \left(\frac{(k-1)(k-2)}{2} - 1\right)\Delta_*.$$

On the other hand, (33) implies

$$(k-1)(K_s - K) \geq (k-1)^2(k-3)x_* + \frac{(k-1)(k-2)}{2}\Delta_*,$$

which dominates the nonconstant terms above. This proves (34). Multiply by $k-2$ and add $(D_k - 1)K$. Since $(k-1)(k-2) = D_k + 1$, the coefficient of K becomes -2 , and (35) follows. $\square$

Let z be the number of non-distinguished terminal portals. Sum (24) over the non-distinguished components and combine the result with (35). This gives

$$(37) \quad (k-2)(k-1)! \leq C_s + (k-2)D_k z.$$

Define

$$(38) \quad Z_s = \max \left\{ 0, \left\lceil \frac{(k-2)(k-1)! - C_s}{(k-2)D_k} \right\rceil \right\}.$$

Then $z \geq Z_s$.

Put

$$(39) \quad E_k = (k-1)(k-3) = D_k - (k-2).$$

Lemma 9.4 (High-entry defect). *Every non-spanning component with $\mu \geq 3$ satisfies*

$$(40) \quad d \geq E_k.$$

Consequently, in layer s , the number of high-minimum-entry non-distinguished components is at most

$$\lfloor K_s / E_k \rfloor.$$

Proof. Apply (21) with $a + b \leq 2$ in (17). One obtains

$$d \geq E_k x + D_k(\mu - 2) + \frac{k-2}{2}\Delta - (k-2).$$

For $\mu \geq 3$ the right-hand side is at least $D_k - (k-2) = E_k$. The count follows from $K \leq K_s$ in Lemma 9.2. $\square$

The source correspondence incurs at most one global portal loss. If no terminal chain is present, the nonterminal source map may omit one final component. If a terminal chain is present, every component is represented by a source, but exactly one source terminates at the terminal symbol. These mutually exclusive cases therefore contribute the same single subtraction of one.

The distinguished component may provide one additional zero-cost target only when it differs from the actual root C_0 . Since ties in the maximum- μ selection are resolved in favor of C_0 , this case implies

$$\mu_{\max} - \mu_{C_0} \geq 1.$$

The exact root term in (14) pays for this additional absorption. Define

$$(41) \quad G_s = \max \left\{ 0, Z_s - 1 - \left\lfloor \frac{K_s}{E_k} \right\rfloor \right\}.$$

Theorem 9.5 (Layerwise correction for reduced paths). *Let $k \geq 5$, let P be a σ^2 -reduced Hamilton path, and let $s = s(P)$ be the canonical layer from Lemma 9.1. Then*

$$(42) \quad \boxed{\text{wt}(P) \geq B_H(k) + s(P) + G_{s(P)}}.$$

Proof. Let z be the number of non-distinguished terminal portals. By the source correspondence in Theorem 4.1, the selected nonterminal sources have codimension at most one in the component set. When $I = 0$, the omitted object may be the unique final component L ; when $I = 1$, every component is a source and the unique terminal source is the sole source without a nonterminal target. The mutual exclusivity of these cases yields one global loss of at most one portal.

If one of those nonterminal chains has zero cost, it has one slot. By Corollary 8.5, its target cannot have minimum entry two. The target map in Theorem 4.1 is injective, including in the self-nested case, so each high-minimum-entry non-distinguished component absorbs at most one zero-cost portal chain. By Lemmas 9.2 and 9.4, there are at most $\lfloor K_s / E_k \rfloor$ such targets.

The distinguished component can be one additional target only when it is not the actual root C_0 , since C_0 is excluded from the target bijection. By the tie convention, this case has $\mu_{\max} - \mu_{C_0} \geq 1$, and the root term in (14) compensates that single additional absorption. Consequently at least

$$z - 1 - \lfloor K_s / E_k \rfloor$$

portal obligations contribute a positive integer either through a chain term or through the compensating root term. Since $z \geq Z_s$, the total correction is at least G_s . Substituting the canonical equality (29) proves (42). $\square$

Corollary 9.6 (Global lower bound). *Define*

$$(43) \quad \Gamma_k = \min_{s \geq 0} (s + G_s).$$

Then, for every $k \geq 5$,

$$(44) \quad \boxed{S(k) \geq \mathcal{H}_k + \Gamma_k.}$$

Moreover,

$$(45) \quad \boxed{\Gamma_k = \left(1 - \frac{3}{k} + O(k^{-2})\right) (k-4)!}.$$

Proof. Choose a minimum-weight Hamilton path. By Theorem 3.4, it has a σ^2 -reduced minimum-weight representative P^* . Apply Theorem 9.5 to the canonical integer layer $s(P^*)$ supplied by Lemma 9.1, and then minimize the objective over all $s \geq 0$. The exact minimizer formula and asymptotic expansion are proved in Appendix C. $\square$

10. NUMERICAL LOWER BOUNDS

All quantities in Equations (27), (36), (38), (41) and (43) are integers. The values in Table 1 are checked in two complementary ways: the Lean theorem `PreimageChain.superperm_numerical_bounds_closed` derives the $S(k)$ inequalities from the unconditional global theorem, while the deterministic Python verifier reconstructs the local transition systems and independently minimizes the exact layer objective.

TABLE 1. Hunter’s certified bound, the revised correction, and the resulting computer-assisted lower bound. Upper bounds are the record 5906 at $k = 7$ and the gain-one construction for $k \geq 8$.

k	$\mathcal{H}_k$	Γ_k	new lower bound	known upper bound
5	153	0	153	153
6	869	0	869	872
7	5,888	4	5,892	5,906
8	46,103	15	46,118	46,204
9	408,338	80	408,418	408,965
10	4,032,575	505	4,033,080	4,037,046
11	43,912,566	3,669	43,916,235	43,948,807
12	522,580,501	30,263	522,610,764	522,910,088
13	6,746,243,919	279,300	6,746,523,219	6,749,568,009
14	93,887,403,944	2,852,497	93,890,256,441	93,924,230,410

For $k = 14$,

$$D_{14} = 155, \quad E_{14} = 143, \quad \kappa_{14} = 27.$$

The minimizing layer is $s = 2,852,497$. At that layer

$$K_s = 442,137,062, \quad Z_s = 3,091,864, \quad \lfloor K_s/E_{14} \rfloor = 3,091,867,$$

so $G_s = 0$. The preceding layer has objective 2,852,506, confirming strict minimality for this value of k .

11. PRIMITIVE LADDERS AND LIMITATIONS OF THE METHOD

11.1. Primitive ladders. For $0 \leq d \leq k-4$, define the *primitive ladder* $T_{k,d}$ to be the component with piece sizes

$$(k-2), (k-1)^{k-3-d}.$$

It has minimum entry two, ends in a full piece, and satisfies

$$t = D_k - (k-1)d, \quad \text{pValue} = A_k - k^2d.$$

Proposition 11.1 (Exact-weight-three primitive antichain). *No $T_{k,d}$ can be followed through an exact weight-three seam by a vertex-disjoint $T_{k,e}$.*

Proof. Put $n = k-1$. Write the source head as $mabR$, where $|R| = n-2$. The classes of its last full piece are

$$S_i = [m\sigma^i(bRa)], \quad i \in \mathbb{Z}_n.$$

An exact weight-three target start is $Rpqt$, where (p, q, t) is a permutation of (m, a, b) . The first partial and first full target pieces contain

$$P_j = [t\sigma^j(Rpq)], \quad F_j = [t\sigma^j(Rqp)].$$

A direct cyclic comparison gives

(p, q, t)	classes already shared with the source
(m, a, b)	$S_0 = F_0, \quad S_{n-1} = P_0$
(m, b, a)	$S_0 = P_{n-2}, \quad S_{n-1} = P_{n-1}$
(a, m, b)	$S_0 = P_0, \quad S_{n-1} = F_0$
(a, b, m)	$S_i = P_{i-1} \quad (i \in \mathbb{Z}_n)$
(b, m, a)	$S_0 = F_{n-2}, \quad S_{n-1} = F_{n-1}$
(b, a, m)	$S_i = F_{i-1} \quad (i \in \mathbb{Z}_n).$

Every orientation therefore shares at least two rotation classes with the source. $\square$

11.2. A two-block bridge family. A *bridge block* B_k is the exact weight-three piece chain

$$(k-3), (k-1)^{k-4}.$$

It has

$$(46) \quad P_k = k^2 - 4k + 1$$

rotation classes, defect $k-1$, minimum entry two, and a full final piece.

Proposition 11.2 (Two-block bridge family). *For every $k \geq 8$, there exists a vertex-disjoint exact weight-three successor B'_k of B_k of the same type. Their union is one strongly exitless component with*

$$t = 2P_k, \quad d = 2(k-1), \quad \pi = 1.$$

Consequently, any universal component inequality of the form

$$(47) \quad t(C) \leq D_k \pi(C) + c_k d(C)$$

must satisfy

$$(48) \quad c_k \geq \frac{k^2 - 5k + 1}{2(k-1)} = \Omega(k).$$

Proof. The explicit successor map and support-disjointness calculation are given in [Appendix B](#). Concatenation adds one weight-three seam and preserves minimum entry two. The combined piece parameters are $r = 2(k - 3)$, $\Delta = 4$, $x = 0$, and $\mu = 2$, giving the displayed t, d, π . Substitution in (47) yields (48). $\square$

Thus the explicit family proves that a universal single-component linear envelope of the specific form (47) cannot have $c_k = O(1)$. It does not exclude nonlinear defect functions, multiple portal states per component, prefix-sensitive potentials, or joint inequalities coupling several components.

11.3. Contraction under neutral refinement.

Proposition 11.3 (Weight-three neutral refinement). *Suppose q internal weight-three edges are cut inside minimum-entry-weight-two components, every resulting block still has minimum entry two, and the global maximum minimum-entry value is unchanged. Then the refined graph has the same value of B_1 . Moreover, the removed edges form a zero-surplus matching saturating every refined block except one per original component.*

Proof. The total edge weight changes by $-3q$, the number of components by $+q$, and the sum of minimum-entry values by $+2q$. Hence B_1 changes by $-3q + q + 2q = 0$. Within each original component, the removed edges join the head of each block to the tail of the next. Relative to the refined baseline, their surplus is $3 - (1 + 2) = 0$, and their sources and targets are distinct. They match all but one block per original component. $\square$

The proposition excludes the derivation of a new factorial level by cutting existing weight-three seams and counting the resulting blocks independently. Any improvement of this type requires higher-order integer packing constraints or a non-neutral second transform.

12. FORMAL VERIFICATION, FINITE AUDITS, AND REPRODUCIBILITY

The immutable ancillary archive accompanying this version has SHA-256

8ce335b549b5c52c4ffbbe91f1a75528ebdbd8d9e3de49d21002c32d14a0dd65.

The archive is versioned as 4.0.1-lean-formalized-release. The formal development uses Lean 4.31.0 and pins the Hunter–Raudvere baseline to

d45222190031d162feb1f6f3cf5fe2d11fab726d.

The public formalization repository is the

[public GitHub repository](#),

and the exact formalization source snapshot is commit

a3852f0de2d84b99cc65d4c281e86892b5f6e911.

The gain-one upper-bound source remains pinned to

62958c6268f728828ed578a171486866b6fe4b20.

Lean theorem boundary. The public root module exports the following final theorems.

Lean theorem	Mathematical role
<code>actual_reduced_pathwise_bound_closed</code>	The corrected lower bound for every σ^2 -reduced Hamilton path.
<code>pathwise_new_bound_closed</code>	The unconditional pathwise theorem, obtained by the non-increasing σ^2 normal form.
<code>superperm_new_bound_closed</code>	The unconditional global lower bound after the Hamilton-path-to-superpermutation bridge.
<code>superperm_numerical_bounds_closed</code>	Direct $S(k)$ inequalities for every $k = 5, \dots, 14$.

The release head was checked with

```
lake --rehash build
lake env lean AxiomCheck.lean
rg -n '\bsorry\b|\badmit\b|\^s*axiom\b' \
  --glob '*.lean' --glob '!vendor/**' --glob '!proof-progress/**' .
```

The rehashed build completed successfully with 8714 jobs, and the active-source scan returned no match. The four exported final theorems use only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`; no project-local axiom is introduced.

The formalization covers the actual component-capacity chain, primitive-block stability, the pointwise actual-image baseline of [Lemma 9.1](#), the root-biased distinguished-component budget, the single-exception capacity-one portal assignment, the σ^2 reduction, the global theorem, and the direct numerical corollaries. The foundational Hunter–Raudvere definitions and theorems are imported from the pinned external library.

Independent deterministic finite audits. A separate deterministic Python audit reproduces the finite transcriptions and certificate checks using only the Python standard library.

File	Purpose
<code>verify_reduced_weight2_normal_form.py</code>	Audits the two weight-two successors and the local monotone-measure identities used in the σ^2 -reduced normal form.
<code>verify_hunter_bound1_sharpness_n6.py</code>	Checks the eight-component order-six certificate: spanning, disjointness, simplicity, strong exitlessness, weights, minimum entry values, and $B_1 = 863$.
<code>verify_preimage_chain_bound_n6.py</code>	Reconstructs $F(P)$, checks the exact identity on nineteen order-four paths, evaluates Π , and calibrates the two order-six examples.
<code>verify_preimage_component_matching.py</code>	Independently reconstructs $F(P)$ and D_P ; exhausts all 720 order-three Hamilton paths and checks archived plus deterministic order-four samples against the source/target component bijections.
<code>verify_full_primitive_block_stability.py</code>	Audits local portal states, primitive transitions, the exact integer layer formula, and the numerical table through order fourteen.
<code>verify_revision_edge_cases.py</code>	Checks the spanning signed-deviation example, the $k = 5$ bridge ambiguity, and the explicit $k = 8, \dots, 14$ bridge construction.
<code>verify_self_nesting_bridge_macro_theory.py</code>	Calibrates self-nesting, terminal-full rigidity, primitive rigidity, and supplementary bridge-macro experiments.

The canonical finite-audit commands are

```
python run_quick_audit.py
python run_full_audit.py
```

The frozen reference environment is Linux `x86_64` with CPython 3.13.5 and no third-party Python packages. Exact commands, claim-to-checker mapping, expected reports, release meta-data, final Lean logs, and SHA-256 manifests are included in the ancillary archive.

Layer	Status
Hunter–Raudvere baseline	Pinned external Lean 4.31.0 development
New arbitrary- k main proof chain	Lean 4.31.0 formalization in the merged release
Order-six certificates and $k \leq 14$ finite computations	Deterministic Python reconstruction and audit
Independent replication	Outside the scope of the present verification record

13. CONCLUSIONS AND OUTLOOK

The preimage-chain identity quantifies the ordering information lost when Hunter components are priced independently. The σ^2 -reduced normal form removes the ambiguity among zero-cost weight-two transitions, while the rate-defect normal form and endpoint-matching capacity theorem convert a bounded defect budget into a family of rigid terminal portals. The pointwise actual-image baseline assigns a canonical nonnegative layer to every Hamilton path, and the capacity-one portal assignment accounts for all but one global source/terminal exception. These ingredients yield the explicit correction Γ_k and the lower bounds in [Table 1](#).

The proof from actual component geometry to the unconditional global $S(k)$ theorem is verified in Lean 4.31.0. The formalization confirms that the single subtraction in G_s is shared by the two mutually exclusive terminal cases and that a distinguished non-root zero-cost target is charged to the exact root gap $\mu_{\max} - \mu_{C_0}$.

The six-symbol analysis distinguishes a low-priced component system from a low-priced Hamilton preimage: the explicit image has $B_1 = 863$, but each of its preimages incurs at least six additional units in the assignment relaxation. This observation does not imply $S(6) \geq 875$, since the known 872-letter construction satisfies $B_1 + \Pi = 866$.

The bridge construction and the neutral-refinement contraction delimit a precise subclass of the method. A single-component, single-portal affine envelope cannot have a bounded defect coefficient, and cutting existing weight-three seams alone creates no new independent obligation. The obstruction does not apply to nonlinear, multistate, intercomponent, prefix-sensitive, or iterated-transform methods. Any improvement beyond the present $\Theta((k-4)!)$ correction therefore requires additional higher-order structure.

AUTHOR CONTRIBUTIONS

Conceptualization, methodology, formal analysis, software, validation, investigation, writing—original draft, and writing—review and editing: Xiaolong Liu.

CONFLICT OF INTEREST

The author declares no conflict of interest.

DATA AND CODE AVAILABILITY

The ancillary release identified in [Section 12](#) contains the deterministic verifiers, compact certificates, canonical reports, exact commands, claim maps, the final Lean release record, selected proof sources, build and axiom-audit logs, and SHA-256 manifests. The formalization is identified by the [public repository](#) and source snapshot `a3852f0de2d84b99cc65d4c281e86892b5f6e911`. The repository also provides the English, Chinese, and bilingual papers and the ancillary archive.

DECLARATION ON GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work, ChatGPT Pro (OpenAI) was used for exploratory derivations, computational prototyping, Lean and Python code development, and language editing. All mathematical claims, proofs, citations, source code, and conclusions were reviewed by Xiaolong Liu, who assumes full responsibility for the published work. The Lean formalization and deterministic computational audits are documented in [Section 12](#); they do not constitute independent peer review.

APPENDIX A. COORDINATE PROOF OF THE PORTAL LEMMAS

Put $n = k - 1$. A piece start is a word $z = z_0 \cdots z_{k-1}$; its marker is z_{k-1} and its cyclic base is $z_0 \cdots z_{k-2}$. A piece of size s contains

$$(49) \quad \mathcal{P}(z, s) = \{[z_{k-1}\sigma^i(z_0 \cdots z_{k-2})] : 0 \leq i < s\}.$$

Its endpoint is $z_{k-1}\sigma^{s-1}(z_0 \cdots z_{k-2})$. An exact weight-three successor retains the suffix after the first three endpoint letters and appends one of their six permutations.

Lemma A.1 (Marked-class alignment). *Let W, W' be permutation words on the same alphabet and let z be a designated symbol. Each rotation class has a unique representative beginning with z . Consequently, $[W] = [W']$ if and only if the representatives obtained by rotating W and W' to place z first are identical letter by letter.*

Proof. The symbol z occurs exactly once in each permutation word, so exactly one cyclic rotation begins with z . Equality of rotation classes is therefore equivalent to equality of these unique representatives. $\square$

A.1. Full runs. Write a full-piece start as Yam , with $|Y| = k - 2$. Direct substitution in (49) shows that its unique support-disjoint full successor is

$$(50) \quad Q(Yam) = \sigma(Y)am.$$

The supports of $z, Qz, \dots, Q^{k-3}z$ are pairwise disjoint. Equality of a class from $Q^i z$ and one from $Q^j z$ aligns the unique letters a, m and then forces $i = j$ modulo $k - 2$. The next support repeats the first. This proves [Lemma 7.1\(i\)](#).

A.2. All six successor orientations at a partial endpoint. Normalize a partial piece of deficit δ so that its endpoint is $mabR$, where $|R| = k - 3$. In the marked complete-two-cycle with marker m and cyclic base abR , label incidences by $\mathbb{Z}_n$ so that the endpoint class is 0. The piece occupies

$$(51) \quad I_\delta = \{0, \delta + 1, \delta + 2, \dots, n - 1\};$$

the omitted cyclic arc is $J_\delta = \{1, \dots, \delta\}$.

The six exact weight-three successor starts are $R\pi(m, a, b)$. If $\pi(m, a, b) = (p, q, t)$, then after j applications of the full-piece map Q the candidate full piece has start

$$\sigma^j(Rp)qt$$

and support

$$(52) \quad \mathcal{F}_j(p, q, t) = \{[t\sigma^\ell(\sigma^j(Rp)q)] : \ell \in \mathbb{Z}_n\}.$$

By [Lemma A.1](#), an intersection can be tested by rotating both words to put the same distinguished letter first. Since the letters of R retain their cyclic order, once the positions of m, a, b are fixed the remaining letters force a unique pair of incidence indices (i, ℓ) . Solving these six

elementary word equalities gives the following complete table. The integer j is the index of the prospective full piece, beginning with $j = 0$; all incidence indices are taken modulo n .

(p, q, t)	$j_{\min}$	one forced equality (i, ℓ)	disjoint full pieces
(m, a, b)	0	$(0, 0)$	0
(m, b, a)	0	$(0, n - 1)$	0
(a, b, m)	0	$(\delta + 1, \delta - 1)$	0
(a, m, b)	$\delta - 1$	$(\delta + 1, n - \delta)$	$\delta - 1$
(b, m, a)	$\delta - 1$	$(\delta + 1, n - \delta - 1)$	$\delta - 1$
(b, a, m)	$n - 2$	$(\delta + 1, \delta)$	$n - 2 = k - 3$.

The notation (i, ℓ) denotes

$$[m\sigma^i(abR)] = [t\sigma^\ell(\sigma^{j_{\min}}(Rp)q)].$$

For the first three orientations the displayed equality already occurs at $j = 0$. For (a, m, b) and (b, m, a) , alignment of the ordered subword R forces $j \geq \delta - 1$, and the displayed pairs show equality at $j = \delta - 1$; thus no earlier full piece intersects. For (b, a, m) , the same alignment forces $j \equiv n - 2 \pmod{n}$, so the candidates $j = 0, \dots, n - 3$ are disjoint and $j = n - 2$ is the first return. These necessity statements exclude every other rotation equivalence, while the displayed equalities prove sufficiency. This proves the endpoint bound in Lemma 7.1(iii).

When $\delta = 1$, the fourth and fifth rows have length zero. The only positive full run uses the last orientation and its boundary full piece contains the unique omitted incidence. A positive full run on the other side of the same unit partial would have to use that same incidence. The two boundary full pieces would overlap, proving Lemma 7.1(ii).

A.3. The internal long-run threshold. Let the left partial deficit be a . Follow the unique long orientation for $n - 2 = k - 3$ full pieces. Normalize the resulting head as $macU$. The six possible right-partial starts are $U\pi(m, a, c)$. Five orientations have an unavoidable common class:

$\pi(m, a, c)$	explicit class already in the accumulated support
(m, a, c)	$[macU]$
(m, c, a)	$[macU]$
(a, m, c)	$[mcUa]$
(a, c, m)	$[mUac]$
(c, m, a)	$[maUc]$.

The remaining orientation is (c, a, m) . Its marker is m . After transporting both boundary pieces to the same marked incidence cycle, choose coordinates so that the left occupied interval is

$$L_a = \{0, a + 1, a + 2, \dots, n - 1\}$$

and the right occupied interval is

$$R_b = \{1, 2, \dots, n - b\}.$$

Thus

$$L_a \cap R_b = \{a + 1, a + 2, \dots, n - b\}, \quad |L_a \cap R_b| = \max(0, n - a - b).$$

The two boundary pieces are disjoint exactly when

$$(53) \quad a + b \geq n = k - 1.$$

The preceding five orientations already have the explicit common classes listed above, so (c, a, m) is the unique orientation that can be disjoint. This proves Lemma 7.1(iv), including both necessity and sufficiency of the threshold.

APPENDIX B. AN EXPLICIT TWO-BLOCK BRIDGE CONSTRUCTION

Assume $k \geq 8$ and put $n = k - 1$. Let a labeled base be $B = (b_0, \dots, b_{n-1})$, with marker m , and set $a = b_{n-3}$. Start with the partial piece at Bm of size $k - 3 = n - 2$. No uniqueness is required for the transition from this partial piece to the first full piece. An explicit choice is made with start

$$(54) \quad z_1 = (b_{n-1}, b_0, \dots, b_{n-4}, b_{n-2}, b_{n-3}, m).$$

After this choice, take the remaining $k - 5$ full pieces successively by the unique full-to-full map Q from (50). The resulting chain, of piece sizes $(k - 3), (k - 1)^{k-4}$, is the bridge block $B_k(B, m)$.

A direct endpoint calculation gives an exact weight-three successor start with the same marker and base

$$(55) \quad \Theta(B) = (b_{n-4}, b_{n-2}, b_{n-1}, b_0, \dots, b_{n-5}, b_{n-3}),$$

where all displayed index ranges are ordinary ranges because $n \geq 7$.

A direct support argument follows. Because the marker m is unique, two classes written with m first are equal exactly when their length- n bases are equal. Delete the anchor $a = b_{n-3}$ from the source bridge cycles. Every source piece is obtained from the fixed cyclic skeleton

$$L = (b_0, b_1, \dots, b_{n-4}, b_{n-2}, b_{n-1})$$

by inserting a in one of its cyclic gaps, except for the single omitted gap

$$(56) \quad (b_{n-5}, b_{n-4}).$$

Indeed, the first partial inserts a between b_{n-4} and b_{n-2} , while the successive full pieces insert it, in order, between

$$(b_{n-2}, b_{n-1}), (b_{n-1}, b_0), (b_0, b_1), \dots, (b_{n-6}, b_{n-5}).$$

For the target base $B' = \Theta(B)$, its varying anchor is $a' = b_{n-6}$. Delete a' and rotate the remaining skeleton to begin with b_0 ; the result is

$$L' = (b_0, \dots, b_{n-7}, b_{n-5}, a, b_{n-4}, b_{n-2}, b_{n-1}).$$

After deleting both a and a' , the source and target have the same common skeleton. In every target bridge cycle the letter a is fixed in the gap (b_{n-5}, b_{n-4}) , because only a' varies among the target gaps. Therefore a target cycle could equal a source cycle only if the source inserted a in the gap (56); that is precisely the one gap absent from the source block. Hence the two bridge supports are disjoint.

The five other permutations of the three endpoint letters already share classes with the source last-full support, as in the six-orientation comparison of Proposition 11.1. Thus (55) supplies the required disjoint successor.

The two blocks together have $r = 2(k - 3)$, $\Delta = 4$, $x = 0$, and $\mu = 2$. Equations (17) and $t = (k - 1)r - \Delta$ give the parameters used in Proposition 11.2.

APPENDIX C. EXACT MINIMIZATION OF THE LAYER CORRECTION

Put

$$M = (k - 1)!, \quad D = D_k, \quad E = (k - 1)(k - 3).$$

Let Z_0 be (38) at $s = 0$. Since $D + 1 = (k - 1)(k - 2)$ and $C_s = C_0 + (D + 1)Ds$,

$$(57) \quad \boxed{Z_s = \max(0, Z_0 - (k - 1)s)}.$$

Also $D = E + (k - 2)$, so

$$(58) \quad \left\lfloor \frac{\kappa_k + sD}{E} \right\rfloor = s + \left\lfloor \frac{\kappa_k + s(k - 2)}{E} \right\rfloor.$$

Consequently

$$(59) \quad G_s = \max \left(0, Z_0 - 1 - ks - \left\lfloor \frac{\kappa_k + s(k - 2)}{E} \right\rfloor \right).$$

Proposition C.1 (Exact minimizing layer). *The value Γ_k is the least integer $s \geq 0$ such that*

$$(60) \quad ks + \left\lfloor \frac{\kappa_k + s(k - 2)}{E} \right\rfloor \geq Z_0 - 1.$$

Proof. While $G_s > 0$, equations (43) and (59) give

$$s + G_s = Z_0 - 1 - (k - 1)s - \left\lfloor \frac{\kappa_k + s(k - 2)}{E} \right\rfloor,$$

which strictly decreases with s . Once $G_s = 0$, the objective equals s and strictly increases. At the preceding layer, failure of (60) is an integer failure by at least one, so its objective is no smaller than the first zero-gap layer. $\square$

Finally,

$$(61) \quad Z_0 = \frac{M}{D} + O(k),$$

because $0 \leq \kappa_k < D$. From (60),

$$\frac{Z_0 - 1 - \kappa_k/E}{k + (k - 2)/E} \leq \Gamma_k \leq \left\lceil \frac{Z_0 - 1}{k} \right\rceil.$$

Thus

$$(62) \quad \Gamma_k = \frac{(k - 1)!}{kD_k} + O(1) = \left(1 - \frac{3}{k} + O(k^{-2}) \right) (k - 4)!,$$

since

$$\frac{(k - 1)!}{kD_k(k - 4)!} = \frac{(k - 1)(k - 2)(k - 3)}{k(k^2 - 3k + 1)} = 1 - \frac{3}{k} + O(k^{-2}).$$

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前像链修正、本原块刚性与超排列的新下界

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摘要

基于 Hunter–Raudvere 已在 Lean 中验证的分量下界框架，本文量化强无提前离环路径分量系统作为某条哈密顿路径的 Hunter 像时所产生的附加代价。前像链分量匹配定理表明：辅助图的路径分量在源端覆盖除一个统一源端/终端例外外的全部 Hunter 分量，在目标端则与全部非根 Hunter 分量形成容量一匹配；存在终端链时需增补一个终端符号。该结构导出精确前像链恒等式，将 $\text{wt}(P) - B_1(F(P))$ 分解为非负的根项、终端项、链项与环项，并给出可计算松弛

$$\text{wt}(P) \geq B_1(F(P)) + \Pi(F(P)).$$

在六符号情形，一个显式 Hunter 像满足 $B_1 = 863$ 、 $\Pi = 6$ ，而已知的 872 字母构造满足 $B_1 = 866$ 、 $\Pi = 0$ 。

本文进一步建立 σ^2 -约化正规形、费率缺陷正规形以及精确权三区间片段链的端点匹配容量定理，并推出本原块稳定性不等式

$$(k-2)t \leq (k-2)D_k\pi + (D_k-1)d, \quad D_k = k^2 - 3k + 1.$$

其中， $\pi = 1$ 当且仅当分量的最小入权为 2 且末片段为满片段。在约化前像中，相应终端门户不能被任何最小入权为 2 的目标零成本吸收，包括自嵌套目标。

固定实际像的点态 Hunter 下界为每条哈密顿路径赋予规范的非负整数层。结合门户刚性、缺陷预算单调性、根优先特选分量包络与容量一目标匹配，可得

$$S(k) \geq \mathcal{H}_k + \Gamma_k, \quad \Gamma_k = \left(1 - \frac{3}{k} + O(k^{-2})\right) (k-4)!.$$

特别地，

$$S(7) \geq 5892, \quad S(8) \geq 46118, \quad S(14) \geq 93,890,256,441.$$

任意 k 的主证明链，包括实际分量容量、点态实际像基线、门户指派、 σ^2 约化、无条件路径级定理、全局 $S(k)$ 定理以及 $k = 5, \dots, 14$ 的数值推论，均已在固定 Hunter–Raudvere 依赖之上用 Lean 4.31.0 形式化。形式化工程不存在未完成证明或项目自定义公理。另有一套仅依赖 Python 标准库的确定性实现，用于重建有限证书、局部转录与精确整数表。

此外, 对 $k \geq 8$ 的显式双桥接块族表明, 形如 $t(C) \leq D_k \pi(C) + c_k d(C)$ 的单分量、单门户线性包络必满足 $c_k = \Omega(k)$ 。这一约束不适用于非线性、多状态、跨分量、前缀敏感或迭代变换方法。

关键词: 超排列; 前像链; 排列重叠图; 哈密顿路径; 形式化验证; Lean 4; 计算机辅助证明; 极值组合数学

数学主题分类 (2020): 05A05, 05C45, 68R15, 68W30

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1 引言

一个 k 符号超排列是定义在 k 字母表上的有限词，并包含该字母表的每个排列作为连续因子。记其最短长度为 $S(k)$ 。目前 $S(1), \dots, S(5)$ 已知，而所有 $k \geq 6$ 的情形仍未解决。Ashlock 与 Tillotson 给出了长度为 $\sum_{i=1}^k i!$ 的经典递归构造 [AT93]；Johnston 证明了原猜想长度下的构造并不唯一 [Joh13]；Houston 则构造了长度为 872 的六符号超排列 [Hou14]。Egan 基于 Williams 在双生成元 Cayley 有向图中的哈密顿路径 [Wil17, Ega18]，得到

$$S(k) \leq k! + (k-1)! + (k-2)! + (k-3)! + k - 3. \quad (1)$$

一项经形式化验证的“增益一” (gain-one) 提升构造对所有 $k \geq 8$ 将该上界进一步降低 1 [Rau26]。本文以 Hunter 与 Raudvere 的下界框架为出发点 [HR26]；该框架的完整定理链已在 Lean 中形式化。记

$$D_k = k^2 - 3k + 1, \quad A_k = k^3 - 2k^2 - k - 1. \quad (2)$$

他们得到的精确点态超排列下界为

$$\mathcal{H}_k = k! + (k-1)! + (k-2)! + k - 3 + \left\lceil \frac{(k-2)! - (k-2)}{D_k} \right\rceil. \quad (3)$$

特别地， $\mathcal{H}_6 = 869$ 、 $\mathcal{H}_7 = 5888$ 、 $\mathcal{H}_8 = 46103$ 。同一形式化工作还刻画了 Hunter 变换的像类：它恰由覆盖全部顶点的强无提前离环路径分量系统组成。这一像刻画将在六符号情形中发挥关键作用。

Hunter 下界对变换后的各分量分别计价。本文关注的核心问题因而不是继续改进单个强无提前离环分量的内部估价，而是：当这些分量必须同时来自同一条哈密顿路径时，还会产生多少不可避免的前像代价。围绕这一问题，本文作出以下贡献。

- (1) 证明前像链分量匹配定理，建立源端与目标端的容量一一对应；据此导出 $\text{wt}(P) - B_1(F(P))$ 的精确恒等式，并定义可计算的最短路一指派松弛 Π 。
- (2) 证明全局 σ^2 -约化正规形，在不增加路径权重的前提下将所有零成本可分解双旋转替换为正规门边。

(3) 建立费率缺陷正规形

$$d = D_k(x + \mu - 2) + (k - 2)\Delta - r, \quad t = D_k m - (k - 1)d.$$

(4) 证明端点匹配容量与本原块稳定性，包括自嵌套终端门户情形。

(5) 证明固定实际像的点态 Hunter 基线，从而为每条哈密顿路径定义规范的非负整数层 $s(P)$ ，而不再把层参数作为隐含假设。

(6) 将终端门户刚性转化为显式分层修正：两个终端情形共用一个全局例外，特选非根目标则由精确根差支付。

(7) 区分 Hunter 真像类与局部松弛分量类，并利用已形式化的像刻画证明二者相等。

(8) 证明两项方法边界：当 $k \geq 8$ 时，双桥接块族强迫所有 $t(C) \leq D_k \pi(C) + c_k d(C)$ 型单分量、单门户线性包络满足 $c_k = \Omega(k)$ ；权三中性细分不产生新的独立块级义务。

下述定理汇总形式化导出的主要数值下界。

定理 1.1 (Lean 形式化数值推论). 无条件形式化定理给出

$$\begin{array}{ll} S(5) \geq 153, & S(6) \geq 869, \\ S(7) \geq 5,892, & S(8) \geq 46,118, \\ S(9) \geq 408,418, & S(10) \geq 4,033,080, \\ S(11) \geq 43,916,235, & S(12) \geq 522,610,764, \\ S(13) \geq 6,746,523,219, & S(14) \geq 93,890,256,441. \end{array}$$

其中 $k = 7, \dots, 14$ 是本文的新下界； $k = 5, 6$ 一并列出，以完整对应形式化导出的数值表。

任意 k 的主证明链已在 Lean 4.31.0 中完成形式化。形式化工程公开于

[GitHub repository](#)。

精确形式化源码快照为提交

`a3852f0de2d84b99cc65d4c281e86892b5f6e911`。

另有一套确定性 Python 实现用于重建有限转录与证书；其功能限于计算审计，不构成第二套定理证明器形式化或外部同行评审。

2 排列重叠图与 Hunter 分量系统

把 $\mathfrak{S}_k$ 的元素写成长为 k 的词。对 $u, v \in \mathfrak{S}_k$, 定义

$$w(u, v) = \min\{d \in \{1, \dots, k\} : u_{d+1} \cdots u_k = v_1 \cdots v_{k-d}\}. \quad (4)$$

对哈密顿路径 $P = (p_1, \dots, p_{k!})$, 令

$$\text{wt}(P) = \sum_{i=1}^{k!-1} w(p_i, p_{i+1}).$$

相应字符串长度为 $k + \text{wt}(P)$, 因此

$$S(k) = L(k) + k, \quad (5)$$

其中 $L(k)$ 是哈密顿路径的最小权重。把词写成 $u = abR$, 其中 $|R| = k - 2$ 。定义

$$\sigma(u) = bRa, \quad \sigma^2(u) = Rab, \quad \tau(u) = Rba. \quad (6)$$

其中 σ 表示权一旋转, τ 表示正规权二门边 (proper weight-two door)。 σ 的轨道称为旋转类; 每类含 k 个顶点, 共有 $(k-1)!$ 个类。若路径在访问当前顶点的权一旋转后继之前, 便沿权至少为 2 的边离开当前旋转类, 则称发生提前离环 (*early exit*)。不发生提前离环的路径称为无提前离环路径 (*exitless path*); 若其末顶点的权一旋转后继也已经出现在路径中, 则称为强无提前离环路径 (*strongly exitless path*)。令真实像类为

$$\mathcal{X}_k = \{F(P) : P \text{ 是哈密顿路径}\}.$$

令 $\mathcal{Y}_k$ 为所有覆盖全体顶点、且每个弱连通分量都是顶点简单强无提前离环路径的有向图。下述结论合并了 Hunter–Raudvere 已形式化的必要性与充分性方向。

定理 2.1 (Hunter–Raudvere 像刻画与分量下界). 当 $k \geq 2$ 时,

$$\boxed{\mathcal{X}_k = \mathcal{Y}_k.}$$

对任意哈密顿路径 P 和 $X = F(P)$:

- (i) X 的每个分量都是完整旋转类的并;
- (ii) 分量 C 的尾 t_C 是 P 在 C 中最早访问的顶点;
- (iii) 令 $\mu_C = \text{minto}(C)$, 则

$$\text{wt}(P) \geq B_1(X) := |\text{Comp}(X)| - 1 + \text{wt}(X) + \sum_C \mu_C - \max_C \mu_C; \quad (7)$$

(iv) 若一个非全覆盖分量含 t 个旋转类, 则

$$\text{pValue}(C) := 1 + \mu_C + \text{wt}(C) \geq \frac{A_k}{D_k} t. \quad (8)$$

此外, 其点态分量定理对每个 $X \in \mathcal{X}_k$ 给出相应的 $B_1(X)$ 下界。

对任意分量, 先定义带符号的费率偏差

$$\delta(C) = D_k \text{pValue}(C) - A_k t(C) \in \mathbb{Z}.$$

若 C 非全覆盖, 则由(8)有 $\delta(C) \geq 0$, 并定义其费率缺陷 (*rate defect*) 为

$$\boxed{d(C) = D_k \text{pValue}(C) - A_k t(C) \geq 0.} \quad (9)$$

全覆盖分量不赋予非负费率缺陷。在“外部为空”的标准约定下, 全覆盖分量的 $\mu(C) = 0$, 其带符号偏差可以为负; 例如经典的五符号、长度 153 的路径满足 $\delta(C) = -17$ 。本文所讨论的每个分量均由一条顶点简单路径实现。若前像链的源分量与目标分量相同, 则称其发生自嵌套; 若被强迫的目标顶点还恰等于该分量的尾, 则称其自闭合。本文将最小入权为二且末个区间片段为满片段的分量称为终端门户 (*terminal portal*)。本原阶梯与桥接块将在第 11 节中定义。

3 σ^2 -约化标准形

(6)中的两类权二后继在门户论证中承担不同作用。下述标准形即为后续论证所需形式。

引理 3.1 (权二后继分类). 设 $k \geq 3$ 且 $u, v \in \mathfrak{S}_k$ 。则 $w(u, v) = 2$ 当且仅当

$$v = \sigma^2 u \quad \text{或} \quad v = \tau u.$$

证明. 写 $u = abR$ 。长度为 $k-2$ 的重叠强迫 v 以 R 开头, 末尾两个字母必须是缺失的 a, b , 且只有两种次序。因此得到 $Rab = \sigma^2 u$ 与 $Rba = \tau u$ 。由于 u 的字母互异, 这两者的权都恰为二。□

定义 3.2. 若哈密顿路径不含边 $u \rightarrow \sigma^2 u$, 则称其为 σ^2 -约化。

对路径 P , 记 $N_\sigma(P)$ 为其中权一旋转边 $u \rightarrow \sigma u$ 的数量。

引理 3.3 (σ^2 手术). 设 $k \geq 3$ 。若哈密顿路径 P 含边 $u \rightarrow \sigma^2 u$, 则存在哈密顿路径 P' , 使

$$\text{wt}(P') \leq \text{wt}(P), \quad N_\sigma(P') \geq N_\sigma(P) + 2.$$

证明. 令 $v = \sigma u$ 、 $w = \sigma^2 u$ 。顶点 v 在 P 中出现于某处。把它从原位置删去, 再插入连续顶点 u, w 之间。由重叠权的三角不等式, 删去一个顶点不会增加路径权; 重新插入该顶点时, 局部边权保持不变, 因为

$$w(u, w) = 2 = w(u, v) + w(v, w) = 1 + 1.$$

新列表是原哈密顿列表的一个排列, 因此仍是哈密顿路径。还需证明严格进展。在原位置, v 的前驱不可能是 u , 因为 u 在路径中的后继已经是 w ; 同样, v 的后继不可能是 w , 因为 w 在路径中的前驱已经是 u 。由于只有 u 的旋转后继是 v , 且 w 是 v 唯一的旋转后继, 删去 v 不会删除任何被 N_σ 计数的边; 重新插入则新增 $u \rightarrow v$ 和 $v \rightarrow w$ 两条旋转边。旧位置的拼接还可能额外产生旋转边, 但不会降低上述净增量。 $\square$

定理 3.4 (σ^2 -约化正规形). 对任意 $k \geq 3$ 的哈密顿路径 P , 存在 σ^2 -约化哈密顿路径 P^* , 满足

$$\text{wt}(P^*) \leq \text{wt}(P).$$

因此, 最小权哈密顿路径可以取为 σ^2 -约化。

证明. 只要仍存在 σ^2 边, 就反复应用引理 3.3。每一步使 N_σ 至少增加二, 而始终有

$$0 \leq N_\sigma(P) \leq k! - 1.$$

故过程必然终止, 且权重从不增加。若初始路径已是极小权, 则终止路径仍具有同一极小权。 $\square$

4 前像链恒等式

令 P 为哈密顿路径, $X = F(P)$ 。Hunter 构造确定一个顶点集 $S(P)$ 。对每个 $v \in S(P)$, 变删除去 P 中离开 v 的边并加入 $v \rightarrow \sigma v$; 若因此产生入度二, 则再删去原先进入重复顶点的路径边。记第一步和第二步删去的边集为 E_1, E_2 , 并令

$$I = \mathbf{1}[p_{k!} \in S(P)].$$

Hunter 的精确权重记账为

$$\text{wt}(P) = \text{wt}(X) + \sum_{e \in E_1} (w(e) - 1) - I + \sum_{e \in E_2} w(e). \quad (10)$$

在 $S(P)$ 上定义有向图 D_P :

$$u \rightarrow_{D_P} v \iff (u, \sigma v) \text{ 是 } P \text{ 的边}. \quad (11)$$

D_P 的每个顶点入度、出度都至多为一, 故其分解为若干有向路径与有向环。记 $C_{-1}(P)$ 为各旋转类中首个被访问顶点的旋转前驱。Hunter 构造的定义性排除条件为

$$S(P) \cap C_{-1}(P) = \emptyset.$$

记 $\mathcal{K}(P)$ 为 D_P 的有向路径分量集。对 $K = (v_1, \dots, v_r) \in \mathcal{K}(P)$, 若 $v_r = p_{k!}$, 则称 K 为终端链。非终端链的源 Hunter 分量与目标 Hunter 分量分别记为 $B(K)$ 与 $A(K)$ 。

定理 4.1 (前像链分量匹配). 设 P 为哈密顿路径, $X = F(P)$, C_0 为 X 中包含 p_1 的根分量。

(i) 每个 $K = (v_1, \dots, v_r) \in \mathcal{K}(P)$ 都有唯一初始边

$$h_{B(K)} \longrightarrow \sigma v_1 \in E_2,$$

其中 $h_{B(K)}$ 是唯一分量 $B(K)$ 的头。

(ii) 每个非终端 K 都有唯一退出边

$$v_r \longrightarrow t_{A(K)} \in E_1,$$

其中 $t_{A(K)}$ 是唯一非根分量 $A(K)$ 的尾。

(iii) 若 $I = 0$, 则不存在终端链。令 L 为包含 $p_{k!}$ 的分量, 则

$$B : \mathcal{K}(P) \xrightarrow{\sim} \text{Comp}(X) \setminus \{L\}, \quad A : \mathcal{K}(P) \xrightarrow{\sim} \text{Comp}(X) \setminus \{C_0\}$$

均为双射。

(iv) 若 $I = 1$, 则恰有一条终端链 $K_\star$, 且

$$B : \mathcal{K}(P) \xrightarrow{\sim} \text{Comp}(X)$$

为双射。规定 $\hat{A}(K_\star) = \star$, 其余链取 $\hat{A}(K) = A(K)$, 则

$$\hat{A} : \mathcal{K}(P) \xrightarrow{\sim} (\text{Comp}(X) \setminus \{C_0\}) \cup \{\star\}$$

为双射。

特别地, 每个 *Hunter* 分量至多是一条前像链的源, 每个非根分量至多是一条非终端前像链的目标。若发生自嵌套, 即 $A(K) = B(K)$, 上述容量为一结论仍然成立。

证明. 令 $K = (v_1, \dots, v_r)$ 为 D_P 的一个路径分量。由于 $v_1 \in S(P)$, 而 $S(P)$ 排除 $C_{-1}(P)$, 故 σv_1 不可能是路径首顶点 p_1 , 因而在 P 中有唯一前驱 b 。 v_1 没有 D_P 前驱, 故 $b \notin S(P)$ 。于是原边 $b \rightarrow \sigma v_1$ 在第一步删除后仍保留; 新增边 $v_1 \rightarrow \sigma v_1$ 使 σv_1 的入度变为二, 原边遂在 E_2 中被删除。又因 $b \notin S(P)$, 没有新增边从 b 发出, 所以 b 是 X 中唯一某分量的头。这证明了 (i) 的存在性与唯一性。

反之, 设 h 是 X 的一个头。若 $h \neq p_{k!}$, 则它在 P 中有唯一后继。若 $h \in S(P)$, 新增旋转边会从 h 发出, 故 $h \notin S(P)$ 。其原路径边不属于 E_1 , 因而只能在 E_2 中被删除; 于是存在唯一 $v \in S(P)$ 使该边为 $h \rightarrow \sigma v$ 。若 v 有 D_P 前驱 $u \in S(P)$, 则 P 中 σv 的前驱唯一性强迫 $u = h$, 与 $h \notin S(P)$ 矛盾。因此 v 是一个 D_P 路径起点。路径末顶点 $p_{k!}$ 是 X 的头, 当且仅当 $p_{k!} \notin S(P)$, 即当且仅当 $I = 0$ 。这证明了 (iii)、(iv) 中的源端双射。

现设 K 非终端, 并令 a 为 v_r 在 P 中的后继。由于 $v_r \in S(P)$, 边 $v_r \rightarrow a$ 属于 E_1 。若存在 $u \in S(P)$ 使 $a = \sigma u$, 则有 $v_r \rightarrow_{D_P} u$, 与 K 的极大性矛盾。因此没有新增旋转边进入 a ; 其唯

一原入边被删除后, a 在 X 中入度为零, 是某分量的尾。又 $a \neq p_1$, 故该分量不是根分量。这证明 (ii)。

反之, 设 $a \neq p_1$ 是 X 的一个尾, 并令 v 为其在 P 中的唯一前驱。边 $v \rightarrow a$ 不可能仅因属于 E_2 而被删除: E_2 删除恰发生在仍有新增旋转边进入目标的情形, 而 a 在 X 中入度为零。因此 $v \rightarrow a \in E_1$, 故 $v \in S(P)$ 。又因为没有新增边进入 a , 不存在 $u \in S(P)$ 使 $a = \sigma u$, 从而 v 是唯一某条 D_P 路径的终点。首顶点 p_1 始终是 X 的尾: 它没有原前驱, 而若有新增边 $v \rightarrow p_1$, 则 $v \in C_{-1}(P) \cap S(P)$, 矛盾。故非终端目标映射恰好双射到 $\text{Comp}(X) \setminus \{C_0\}$ 。

最后, $p_k!$ 属于 D_P 当且仅当 $I = 1$ 。此时它没有 D_P 后继, 且位于唯一终端路径分量中; 当 $I = 0$ 时不存在终端链。有向环既无起点也无终点, 只对应 E_1 的内部边, 不改变源端或目标端匹配。定理得证。□

若 $c = |\text{Comp}(X)|$, 该定理给出恰好 $c - 1$ 条非终端链及 $c - 1 + I$ 条总链。有向环不具有分量级源端或目标端, 只贡献内部 E_1 边。

对非终端链定义

$$g(K) = [w(h_{B(K)}, \sigma v_1) - 2] + \sum_{i=1}^{r-1} [w(v_i, \sigma v_{i+1}) - 1] + [w(v_r, t_{A(K)}) - \mu_{A(K)}]. \quad (12)$$

对终端链删去最后一个括号。对有向环 $Z = (v_1, \dots, v_r)$, 令

$$g(Z) = \sum_{i=1}^r [w(v_i, \sigma v_{i+1}) - 1], \quad v_{r+1} = v_1. \quad (13)$$

上述各括号中的量均为非负整数。

定理 4.2 (前像链精确恒等式). 令 C_0 为 X 中包含 P 首顶点的分量, 并令 $\mu_{\max} = \max_C \mu_C$ 。则

$$\boxed{\text{wt}(P) - B_1(X) = \mu_{\max} - \mu_{C_0} + I + \sum_K g(K) + \sum_Z g(Z)}. \quad (14)$$

证明. 由定理 4.1, D_P 各路径分量的初始边恰为 E_2 的分量边界边, 数量为 $c - 1 + I$; 非终端退出边则与所有 $A \neq C_0$ 的分量一一对应。把 E_1 分为链内部边、上述非终端退出边与环边。对每条 E_2 边抽取基线二, 对进入每个 $A \neq C_0$ 的唯一退出边抽取基线 $\mu_A - 1$ 。连同(10)中的 $-I$, 抽取的基线为

$$2(c - 1 + I) + \sum_{A \neq C_0} (\mu_A - 1) - I.$$

它等于

$$(c - 1) + \sum_A \mu_A - \mu_{\max} + (\mu_{\max} - \mu_{C_0}) + I.$$

前三项与 $\text{wt}(X)$ 合并正是 $B_1(X)$; 其余边项恰为(12)与(13)中的量。有向环只贡献 $g(Z)$, 不产生分量边界项。□

4.1 最短路—指派松弛

令

$$\Omega_X = \{v : (v, \sigma v) \in X\}$$

为旋转槽位集。对分量头 B 、不同槽位 $u \neq v$ 和分量尾 A ，定义

$$a_B(v) = w(h_B, \sigma v) - 2, \quad b(u, v) = w(u, \sigma v) - 1,$$

$$c_A(v) = w(v, t_A) - \mu_A, \quad v \notin A.$$

它们都是非负整数。以下在扩展非负整数

$$\overline{\mathbb{N}} = \mathbb{N} \cup \{+\infty\}$$

中工作。令 $d_X(B, A)$ 为所有非空槽位序列中下式的最小值：

$$a_B(v_1) + \sum_{i < r} b(v_i, v_{i+1}) + c_A(v_r),$$

并要求 $v_r \notin A$ ；若不存在这样的序列，规定 $d_X(B, A) = +\infty$ 。类似定义 $d_X(B, \star)$ ，但删去退出项。当 $I = 0$ 时，选择候选根 R 、候选末分量 L 和双射

$$\phi : \text{Comp}(X) \setminus \{L\} \longrightarrow \text{Comp}(X) \setminus \{R\}.$$

令

$$\Pi_0(X) = \min \left(\mu_{\max} - \mu_R + \sum_{B \neq L} d_X(B, \phi(B)) \right).$$

当 $I = 1$ 时，选择 R 和双射

$$\phi : \text{Comp}(X) \longrightarrow (\text{Comp}(X) \setminus \{R\}) \cup \{\star\},$$

并令

$$\Pi_1(X) = 1 + \min \left(\mu_{\max} - \mu_R + \sum_B d_X(B, \phi(B)) \right).$$

最后定义 $\Pi(X) = \min(\Pi_0(X), \Pi_1(X))$ 。

推论 4.3 (前像链加强). 对任意哈密顿路径 P ,

$$\text{wt}(P) \geq B_1(F(P)) + \Pi(F(P)).$$

因此 $B_1 + \Pi$ 逐点支配 B_1 ；它在某些显式像上严格改进，但在记录构造上也可能取等。

证明. 当 $I = 0$ 时, 定理 4.1 在源端给出到 $\text{Comp}(X) \setminus \{L\}$ 的双射, 在目标端给出到 $\text{Comp}(X) \setminus \{C_0\}$ 的双射；复合二者即得到 Π_0 定义中的一个可行指派。当 $I = 1$ 时, 同一定理给出到 $(\text{Comp}(X) \setminus \{C_0\}) \cup \{\star\}$ 的可行指派。真实槽位序列是相应 d_X 项的可行路径。把每条路径替换为独立最短路径、允许不同路径重复使用槽位, 并删去非负的环代价, 只会降低(14)的右端。□

5 六符号情形：锐性与前像修正

随文证书由八条两两不交的强无提前离环路径组成，并覆盖 $\mathfrak{S}_6$ ；其分量大小为

$$4 \times 114, \quad 2 \times 108, \quad 2 \times 24.$$

权重与最小入权为

数量	$ C $	$\text{wt}(C)$	μ_C
4	114	134	2
2	108	127	2
2	24	26	2

因此

$$B_1(X) = 7 + 842 + (16 - 2) = 863.$$

验证器检查覆盖性、不交性、简单性和强无提前离环性，所以 $X \in \mathcal{Y}_6$ 。由定理 2.1 中已形式化的充分性方向， $X \in \mathcal{X}_6$ ，因而至少有一个哈密顿前像。Hunter 的点态分量定理对每个 $Y \in \mathcal{X}_6$ 给出 $B_1(Y) \geq 863$ 。故

$$\boxed{\min_{Y \in \mathcal{X}_6} B_1(Y) = 863.} \quad (15)$$

对这一固定像，最短路—指派松弛给出

$$\Pi_0(X) = 6, \quad \Pi_1(X) = 7.$$

所以任何满足 $F(P) = X$ 的哈密顿路径都有权至少 869，由该前像生成的字符串长度至少为 875。这只是针对固定像 X 的结论，不是全局断言 $S(6) \geq 875$ 。相反，一个已知的长度为 872 的字符串具有路径权 866、分量规模分布

$$25 \times 24 + 1 \times 120,$$

且

$$B_1(F(P)) = 866, \quad \Pi(F(P)) = 0.$$

因此，新修正排除了一个由分量独立计价产生的伪极值，同时在已知记录构造上保持精确取等。

6 费率缺陷与区间片段几何

以下均设 $k \geq 5$ ，且除非特别说明， C 为非全覆盖强无提前离环分量。该分量会连续、完整地访问每个旋转类。在约化分量中，相邻旋转类之间的权二边是正规权二门边；由这类门边连接的极大游程称为区间片段 (*interval piece*)。记：

- $t = t(C)$ 为旋转类数；
- $r = r(C)$ 为区间片段数；

- $\Delta = (k-1)r - t$ 为总片段亏缺；
- x 为所有片段接缝的“接缝权减三”之和；
- $\mu = \text{minto}(C)$ 。

若一个片段含 $k-1$ 个类，则称其为 满片段；否则称为非满片段。每个旋转类贡献 $k-1$ 条旋转边，片段内部的 $t-r$ 条门边权为二，而 $r-1$ 条接缝总权为 $3(r-1) + x$ 。所以

$$\text{wt}(C) = (k+1)t + r + x - 3, \quad \text{pValue}(C) = (k+1)t + r + x + \mu - 2. \quad (16)$$

命题 6.1 (费率缺陷标准形). 设 C 非全覆盖，并令 $d = d(C)$ 。则

$$d = D_k(x + \mu - 2) + (k-2)\Delta - r. \quad (17)$$

若

$$m = \Delta + (k-1)(x + \mu - 2), \quad (18)$$

则

$$t = D_k m - (k-1)d, \quad \text{pValue}(C) = A_k m - k^2 d. \quad (19)$$

特别地，

$$t \equiv -(k-1)d \pmod{D_k}.$$

证明. 把(16)代入 $d = D_k \text{pValue} - A_k t$ ，并使用

$$A_k = (k+1)D_k + k - 2, \quad D_k = (k-1)(k-2) - 1, \quad t = (k-1)r - \Delta.$$

解所得线性方程即得(19)。 □

7 区间片段链的门户游程容量

一条精确权三片段链是由若干区间片段构成的序列，其支撑两两不交，且相邻片段之间的接缝权恰为三。非满片段的亏缺记为 $\delta \geq 1$ 。

引理 7.1 (门户游程局部几何). 在一条精确权三链中：

- (i) 纯满片段游程的长度至多为 $k-2$ ；
- (ii) 亏缺为一的非满片段不可能同时在左右两侧邻接正长度满片段游程；
- (iii) 与非满片段邻接的端点满片段游程长度至多为 $k-3$ ；
- (iv) 内部满片段游程长度通常至多为 $k-4$ ；只有当两侧亏缺量 a, b 满足 $a + b \geq k-1$ 时，长度 $k-3$ 才可能出现。

适用于任意 k 的完整坐标证明见附录 A；其中列出六种精确权三后继方向，并证明循环遗漏弧引理。有限程序仅用于核验上述坐标推导的转录与实现。设一条链含 n_1 个亏缺为一的非满片段，以及 n_2 个亏缺至少为二的非满片段。令 $e \in \{0, 1, 2\}$ 表示链的两个端点中由满片段占据的个数， R 为正长度满片段游程的数量， F 为满片段的数量， N 为长度达到 $k-3$ 的内部游程数量。

定理 7.2 (端点匹配容量). 对任意含 r 个片段、总亏缺为 Δ 的精确权三链，

$$2r \leq (k-2)(\Delta + e). \quad (20)$$

证明. 亏缺为一的门户的单侧性给出

$$2R - e \leq n_1 + 2n_2 \leq \Delta,$$

故 $R \leq (\Delta + e)/2$ 。每个满片段游程的基线长度至多为 $k-4$ ；端点游程与 N 个长内部游程各可额外贡献一个满片段。因此

$$F \leq (k-4)R + e + N.$$

令 $p = n_1 + n_2$ ，并记 $E = \Delta - p = \sum(\delta_i - 1)$ 。每个长内部游程至少消耗 $k-3$ 个相邻亏缺超额单位，而每个单位最多被相邻两条游程计数，所以 $(k-3)N \leq 2E$ 。于是

$$r = p + F \leq \Delta - E + \frac{k-4}{2}(\Delta + e) + e + \frac{2E}{k-3}.$$

从 $(k-2)(\Delta + e)/2$ 中减去右端，剩余 $E(k-5)/(k-3) \geq 0$ 。纯满片段情形有 $\Delta = 0$ 、 $e = 2$ ，且由引理 7.1 知 $r \leq k-2$ 。□

若一个分量的接缝余量总和为 x ，就在所有接缝余量为正的接缝处切开该分量。若 $a, b \in \{0, 1\}$ 分别表示其第一个与最后一个片段是否为满片段，则所得各链的满片段端点总数至多为 $2x + a + b$ 。对(20)求和，得到

$$2r \leq (k-2)(\Delta + 2x + a + b). \quad (21)$$

8 本原块稳定性与刚性终端门户

定义 8.1 (终端本原块证书). 对分量 C ，定义

$$\pi(C) = \begin{cases} 1, & \mu(C) = 2 \text{ 且 } C \text{ 的末个区间片段为满片段,} \\ 0, & \text{其他情况.} \end{cases} \quad (22)$$

当 $\pi(C) = 1$ 时，末个满片段给出一个大小为 D_k 的免费仿射容量证书；本文称该证书为终端本原块。该对象是附着于分量的仿射证书，而不是通过切分路径产生的新弱连通分量。

定理 8.2 (本原块稳定性). 对任意非全覆盖的强无提前离环分量 C ,

$$2d(C) \geq (k-2)(m(C) - \pi(C)). \quad (23)$$

等价地,

$$(k-2)t(C) \leq (k-2)D_k\pi(C) + (D_k-1)d(C). \quad (24)$$

证明. 由(17)、(18)和(21),

$$\begin{aligned} 2d - (k-2)m &= (D_k-1)(x + \mu - 2) + (k-2)\Delta - 2r \\ &\geq (k-1)(k-4)x + (D_k-1)(\mu-2) - (k-2)(a+b). \end{aligned}$$

若 $\mu = 2$, 第一个片段必为非满片段: 若首片段为满片段, 则其尾的两类权二前驱所在的旋转类都已包含在该片段中, 与 $\mu = 2$ 矛盾. 因此 $a = 0$ 、 $b = \pi$, 右端至少为 $-(k-2)\pi$. 若 $\mu \geq 3$, 则 $\pi = 0$ 且 $a + b \leq 2$; 最坏情形 $x = 0$ 、 $\mu = 3$ 、 $a + b = 2$ 时, 剩余

$$(D_k-1) - 2(k-2) = (k-1)(k-4) \geq 0.$$

这证明(23); 再代入 $t = D_k m - (k-1)d$ 即得(24). $\square$

8.1 约化路径中的零成本链

令 P 为 σ^2 -约化路径, C 为 $F(P)$ 的一个分量, 头为 h_C . 零成本前像链至多包含一个槽位, 因为任意两个不同槽位之间的内部转移成本均至少为一。

引理 8.3 (约化路径中的零成本二分). 设一条零成本单槽位链离开 C , 并进入满足 $\mu_A = 2$ 的分量 A . 则槽位与目标尾被唯一强迫为

$$q = \sigma^{-1}\tau(h_C), \quad t_A = \Psi(h_C) := \tau(\sigma^{-1}\tau(h_C)). \quad (25)$$

若 $\Psi(h_C)$ 的旋转类属于 C , 则要么 $A = C$ 且 $t_C = \Psi(h_C)$, 要么这样的零成本链不存在。

证明. 零进入成本给出 $w(h_C, \sigma q) = 2$. 由引理 3.1, σq 要么为 $\sigma^2 h_C$, 要么为 τh_C . 第一种会成为约化路径 P 中的边, 已被排除; 故 $q = \sigma^{-1}\tau h_C$. 零退出成本给出 $w(q, t_A) = 2$. 同样的分类与约化性排除 $t_A = \sigma^2 q$, 只剩 $t_A = \tau q = \Psi(h_C)$. 若 $A \neq C$, 则 A 包含其尾所在的完整旋转类, 而该旋转类已经包含于 C , 与分量不交矛盾. 若 $A = C$, 旋转类相同仍不够: 实际目标顶点必须等于分量尾, 即 $t_C = \Psi(h_C)$. $\square$

引理 8.4 (终端满片段刚性). 最后一个片段为满片段的简单分量不可能满足 $t_C = \Psi(h_C)$.

证明. 把未满片段的起始词写作 Cm , 其中 m 为标记字母、 $|C| = k-1$. 其头为 $m\sigma^{-1}(C)$. 代入(25)得

$$\Psi(h_C) = \sigma(C)m.$$

未满片段包含旋转类 $[m\sigma(C)]$, 并访问该类的每一个具体顶点, 包括 $\sigma(C)m$. 若该顶点同时又是分量尾, 路径将重复顶点. $\square$

推论 8.5 (刚性终端门户). 在 σ^2 -约化哈密顿路径中, 被 $\pi(C)$ 计数的分量不能以零前像链成本进入任何最小入权为二的分量, 包括自身。零成本仅可能出现在两种情形: 该分量本身成为全局末分量, 或其目标分量的最小入权至少为 3。

9 分层下界定理

令

$$B_H(k) = \mathcal{H}_k - k \quad (26)$$

为 Hunter 路径下界, 并定义

$$\kappa_k = D_k \left\lceil \frac{(k-2)! - (k-2)}{D_k} \right\rceil - ((k-2)! - (k-2)), \quad 0 \leq \kappa_k < D_k. \quad (27)$$

引理 9.1 (固定实际像的点态基线). 对任意 $k \geq 5$ 和任意哈密顿路径 P ,

$$B_1(F(P)) \geq B_H(k). \quad (28)$$

因此

$$s(P) := B_1(F(P)) - B_H(k) \in \mathbb{N}, \quad B_1(F(P)) = B_H(k) + s(P). \quad (29)$$

证明. 具体像 $F(P)$ 属于真实像类 $\mathcal{X}_k$, 故定理 2.1 中的点态 Hunter–Raudvere 降维可直接作用于该像。代入长度为 $k(k-1)(k-2)$ 的枢轴强无提前离环路径, 得到

$$B_1(F(P)) \geq k! + (k-1)! + (k-2)! - 3 + \frac{(k-2)! - (k-2)}{D_k}.$$

左端为整数, 对最后的分式取上整, 即得到(28), 因为所得右端恰为 $\mathcal{H}_k - k$ 。在自然数中作差即得(29)。□

本节以下固定一条哈密顿路径 P , 令 $X = F(P)$ 、 $s = s(P)$, 并定义

$$K_s = \kappa_k + sD_k. \quad (30)$$

令 C_0 为包含路径首顶点的真实根分量。选择一个 μ 最大的分量 C^* 作为特选分量; 若真实根与最大值并列, 则选择 C_0 。若 X 只有一个分量, 规定 $K = 0$; 否则每个非特选分量都必为非全覆盖分量, 令 K 为这些分量的非负费率缺陷之和。把 C^* 的参数记为 t_*, r_*, Δ_*, x_* 。代入该层等式得到

$$K_s - K = D_k(r_* + x_*) - (k-2)(t_* - 1). \quad (31)$$

引理 9.2 (缺陷预算单调性). 对任意分层分解,

$$0 \leq K \leq K_s. \quad (32)$$

更精确地,

$$K_s - K \geq (k-1)(k-3)x_* + \frac{k-2}{2}\Delta_*. \quad (33)$$

证明. 由 $t_\star = (k-1)r_\star - \Delta_\star$, 把(31)改写为

$$K_s - K = D_k x_\star + (k-2)\Delta_\star - r_\star + (k-2).$$

由(21)与 $a+b \leq 2$,

$$r_\star \leq \frac{k-2}{2}\Delta_\star + (k-2)x_\star + (k-2).$$

代入即得(33)。又因 K 是非负缺陷之和, 故(32)成立。 $\square$

定理 9.3 (特选分量包络). 对任意层 $s \geq 0$,

$$t_\star \leq (k-1)(k-2) + (k-1)(K_s - K), \quad (34)$$

且

$$(k-2)t_\star + (D_k - 1)K \leq C_s, \quad (35)$$

其中

$$C_s = (k-1)(k-2)^2 + (D_k + 1)K_s. \quad (36)$$

证明. 分量容量给出

$$t_\star = (k-1)r_\star - \Delta_\star \leq (k-1)(k-2) + (k-1)(k-2)x_\star + \left(\frac{(k-1)(k-2)}{2} - 1\right)\Delta_\star.$$

另一方面, (33)意味着

$$(k-1)(K_s - K) \geq (k-1)^2(k-3)x_\star + \frac{(k-1)(k-2)}{2}\Delta_\star,$$

它支配前式的所有非常数项, 从而得到(34)。将其乘以 $k-2$, 再加上 $(D_k-1)K$ 。利用 $(k-1)(k-2) = D_k + 1$, K 的系数化为 -2 , 于是得到(35)。 $\square$

令 z 为非特选分量所含终端门户的数量。对非特选分量求和(24), 再与(35)合并, 得到

$$(k-2)(k-1)! \leq C_s + (k-2)D_k z. \quad (37)$$

定义

$$Z_s = \max \left\{ 0, \left\lceil \frac{(k-2)(k-1)! - C_s}{(k-2)D_k} \right\rceil \right\}. \quad (38)$$

于是 $z \geq Z_s$ 。令

$$E_k = (k-1)(k-3) = D_k - (k-2). \quad (39)$$

引理 9.4 (高最小入权分量的缺陷下界). 任意满足 $\mu \geq 3$ 的非全覆盖分量都有

$$d \geq E_k. \quad (40)$$

所以在层 s 中, 最小入权至少为三的非特选分量数量至多为

$$\lfloor K_s / E_k \rfloor.$$

证明. 把(21)中的 $a + b \leq 2$ 代入(17), 得到

$$d \geq E_k x + D_k(\mu - 2) + \frac{k-2}{2}\Delta - (k-2).$$

当 $\mu \geq 3$ 时, 右端至少为 $D_k - (k-2) = E_k$. 数量结论由引理 9.2 中的 $K \leq K_s$ 得到. $\square$

源端对应在全局至多损失一个门户义务. 若不存在终端链, 则非终端源映射可能遗漏一个全局末分量; 若存在终端链, 则每个分量均有对应源, 但恰有一个源以终端符号为目标. 两种情形互斥, 因而总账中仅出现一个减一项.

另一个可能的零成本目标是特选分量. 它只有在不是真实根 C_0 时才会成为额外目标. 由于最大 μ 并列时优先选择 C_0 , 此时必有

$$\mu_{\max} - \mu_{C_0} \geq 1.$$

因此, (14) 中的精确根项恰好支付这一次额外吸收. 定义

$$G_s = \max \left\{ 0, Z_s - 1 - \left\lfloor \frac{K_s}{E_k} \right\rfloor \right\}. \quad (41)$$

定理 9.5 (约化路径的分层修正定理). 设 $k \geq 5$, P 是 σ^2 -约化哈密顿路径, 且 $s = s(P)$ 为引理 9.1 给出的规范层. 则

$$\boxed{\text{wt}(P) \geq B_H(k) + s(P) + G_{s(P)}}. \quad (42)$$

证明. 由定理 4.1 的源端对应, 实际非终端链源在分量集合中的余维至多为一. 当 $I = 0$ 时, 缺失者可以是唯一全局末分量; 当 $I = 1$ 时, 每个分量均为链源, 而唯一终端源是唯一没有非终端目标的源. 两种情形互斥, 因此门户总账中至多损失一个单位.

若由终端门户发出的链总成本为零, 则它只含一个槽位. 由推论 8.5, 其目标不可能具有最小入权为二. 定理 4.1 的目标端映射是单射, 所以每个最小入权至少为三的非特选分量至多零成本吸收一条门户链; 自嵌套只对应 $A(K) = B(K)$, 并不破坏该容量为一性质. 由引理 9.4, 这类目标至多有 $\lfloor K_s/E_k \rfloor$ 个.

若特选分量还作为额外零成本目标, 则它不是根分量 C_0 . 由最大最小入权并列时选择根分量的约定, 根修正 $\mu_{\max} - \mu_{C_0}$ 至少为一, 恰好补偿这一额外吸收. 因此, 至少有

$$Z_s - 1 - \lfloor K_s/E_k \rfloor$$

条由终端门户发出的链在(14)中贡献正整数; 取非负部分得到 G_s ; 再代入规范层等式(29), 即得(42). $\square$

推论 9.6 (全局下界). 定义

$$\Gamma_k = \min_{s \geq 0} (s + G_s). \quad (43)$$

则对任意 $k \geq 5$,

$$\boxed{S(k) \geq \mathcal{H}_k + \Gamma_k}. \quad (44)$$

并且

$$\Gamma_k = \left(1 - \frac{3}{k} + O(k^{-2})\right) (k-4)!. \quad (45)$$

证明. 选择一条最小权哈密顿路径。由定理 3.4, 存在一条具有同样最小权的 σ^2 -约化代表 P^* 。对引理 9.1 给出的规范整数层 $s(P^*)$ 应用定理 9.5, 再对所有 $s \geq 0$ 的目标值取最小。精确最优层公式与渐近展开见附录 C。□

10 数值下界

式 (27)、(36)、(38)、(41) 与 (43) 中的量全为整数。表 1 中的数值由两条互补路线核验: Lean 定理 `PreimageChain.superperm_numerical_bounds_closed` 从无条件全局定理直接推出 $S(k)$ 不等式; 确定性 Python 程序则从头重建局部转移系统, 并独立最小化精确层目标。

表 1: Hunter–Raudvere 已形式化验证的下界、本文分层修正项及所得计算机辅助下界。 $k = 7$ 的上界为记录值 5906; $k \geq 8$ 使用“增益一”构造。

k	$\mathcal{H}_k$	Γ_k	本文下界	当前上界
5	153	0	153	153
6	869	0	869	872
7	5,888	4	5,892	5,906
8	46,103	15	46,118	46,204
9	408,338	80	408,418	408,965
10	4,032,575	505	4,033,080	4,037,046
11	43,912,566	3,669	43,916,235	43,948,807
12	522,580,501	30,263	522,610,764	522,910,088
13	6,746,243,919	279,300	6,746,523,219	6,749,568,009
14	93,887,403,944	2,852,497	93,890,256,441	93,924,230,410

当 $k = 14$ 时,

$$D_{14} = 155, \quad E_{14} = 143, \quad \kappa_{14} = 27.$$

最优层为 $s = 2,852,497$ 。在该层,

$$K_s = 442,137,062, \quad Z_s = 3,091,864, \quad \lfloor K_s/E_{14} \rfloor = 3,091,867,$$

所以 $G_s = 0$ 。前一层的目标值为 2,852,506, 从而严格确认该整数最小值。

11 本原阶梯与方法边界

11.1 本原阶梯

对 $0 \leq d \leq k-4$, 定义本原阶梯 $T_{k,d}$ 为片段大小序列

$$(k-2), (k-1)^{k-3-d}$$

所确定的分量。它的最小入权为二, 以满片段结束, 并满足

$$t = D_k - (k-1)d, \quad \text{pValue} = A_k - k^2d.$$

命题 11.1 (精确权三本原反链). 不存在通过一条精确权三接缝把 $T_{k,d}$ 接到一个与其顶点不交的 $T_{k,e}$ 的方式。

证明. 令 $n = k-1$. 把源分量的头写为 $mabR$, 其中 $|R| = n-2$. 其末满片段的旋转类为

$$S_i = [m\sigma^i(bRa)], \quad i \in \mathbb{Z}_n.$$

任意精确权三目标起始词可写为 $Rpqt$, 其中 (p, q, t) 是 (m, a, b) 的一个排列。目标的第一个非满片段与首个满片段分别含

$$P_j = [t\sigma^j(Rpq)], \quad F_j = [t\sigma^j(Rqp)].$$

直接比较循环词得到

(p, q, t)	与源组件已经共享的旋转类
(m, a, b)	$S_0 = F_0, \quad S_{n-1} = P_0$
(m, b, a)	$S_0 = P_{n-2}, \quad S_{n-1} = P_{n-1}$
(a, m, b)	$S_0 = P_0, \quad S_{n-1} = F_0$
(a, b, m)	$S_i = P_{i-1} \quad (i \in \mathbb{Z}_n)$
(b, m, a)	$S_0 = F_{n-2}, \quad S_{n-1} = F_{n-1}$
(b, a, m)	$S_i = F_{i-1} \quad (i \in \mathbb{Z}_n).$

每一种方向都至少与源分量共享两个旋转类。 □

11.2 双桥接块族

定义桥接块 B_k 为精确权三片段链

$$(k-3), (k-1)^{k-4}.$$

它覆盖

$$P_k = k^2 - 4k + 1 \tag{46}$$

个旋转类, defect 为 $k-1$, 最小入权为二, 并以满片段结束。

命题 11.2 (双桥接块族). 对每个 $k \geq 8$, 存在一个与 B_k 顶点不交、并通过精确权三接缝接续的同型桥接块 B'_k 。二者的并是一条强无提前离环分量, 且

$$t = 2P_k, \quad d = 2(k-1), \quad \pi = 1.$$

因此, 任意形如

$$t(C) \leq D_k \pi(C) + c_k d(C) \quad (47)$$

的普适分量不等式都必须满足

$$c_k \geq \frac{k^2 - 5k + 1}{2(k-1)} = \Omega(k). \quad (48)$$

证明. 显式后继映射和支撑不交计算见附录 B。拼接只增加一条权三接缝, 并保持最小入权为二。合并后的参数为 $r = 2(k-3)$ 、 $\Delta = 4$ 、 $x = 0$ 、 $\mu = 2$, 从而得到所列 t, d, π 。代入(47)即得(48)。□

因此, 命题 11.2 只对精确形式(47)作出限制: 任何这样的单分量、单门户线性包络都必须具有 $c_k = \Omega(k)$ 。该命题并不排除含附加常数、非线性缺陷函数、多门户状态、跨分量势函数、前缀信息或迭代 Hunter 变换的方法。

11.3 中性细分的收缩

命题 11.3 (权三中性细分). 设在若干最小入权为二的分量内部切开 q 条权三边; 切开后的每个块仍有最小入权为 2, 且全局最大最小入权不变。则细分后的图的 B_1 与原图相同。此外, 被删去的边形成一个零超额匹配: 在每个原分量内, 除一个细分块外, 其余块都被匹配饱和。

证明. 总边权改变 $-3q$, 分量数改变 $+q$, 最小入权之和改变 $+2q$, 所以 B_1 的变化为 $-3q + q + 2q = 0$ 。在每个原分量内, 被删去的边连接每个块的头与下一块的尾; 相对细化后的基线, 其超额为 $3 - (1 + 2) = 0$, 且所有源、目标都互异。因此它们匹配掉每个原分量中除一个块外的所有块。□

该命题表明, 仅沿既有权三接缝切分分量并独立计数所得块, 不能产生新的阶乘层级。此类改进需要更高阶的整数打包约束, 或非中性的二级变换。

12 形式化验证、有限审计与可复现性

本版冻结补充材料归档的 SHA-256 为

8ce335b549b5c52c4ffbbe91f1a75528ebdbd8d9e3de49d21002c32d14a0dd65。

版本号 4.0.1-lean-formalized-release。形式化工程使用 Lean 4.31.0。Hunter-Raudvere 基线固定在

d45222190031d162feb1f6f3cf5fe2d11fab726d。

公开形式化仓库为

[GitHub repository](#),

本文引用的精确形式化源码快照为提交

a3852f0de2d84b99cc65d4c281e86892b5f6e911。

“增益一”上界源码仍固定在

62958c6268f728828ed578a171486866b6fe4b20。

Lean 定理边界

公共根模块导出下列最终定理。

Lean 定理	数学作用
actual_reduced_pathwise_bound_closed	每条 σ^2 -约化哈密顿路径的修正下界。
pathwise_new_bound_closed	通过不增权的 σ^2 约化正规形得到无条件路径级定理。
superperm_new_bound_closed	经哈密顿路径-超排列桥接得到无条件全局下界。
superperm_numerical_bounds_closed	对 $k = 5, \dots, 14$ 直接给出正式 $S(k)$ 数值不等式。

发布头使用以下命令核验：

```
lake --rehash build
lake env lean AxiomCheck.lean
rg -n '\bsorry\b|\badmit\b|^\s*axiom\b' \
  --glob '*.lean' --glob '!vendor/**' --glob '!proof-progress/**' .
```

重哈希构建成功完成 8714 个任务；活动源码扫描为空。四个最终导出定理仅依赖标准逻辑公理 `propext`、`Classical.choice` 与 `Quot.sound`，未引入项目自定义公理。

形式化覆盖实际分量容量、本原块稳定性、引理 9.1 的点态实际像基线、根优先特选分量预算、单一例外的容量一门户指派、 σ^2 约化、全局定理与直接数值推论。基础 Hunter–Raudvere 定义和定理由固定的外部库导入。

确定性有限审计

一套独立实现的确定性 Python 审计仅依赖 Python 标准库，用于重建有限转录与证书检查。

文件	用途
verify_reduced_weight2_normal_form.py	审计两种权二后继，以及 σ^2 -约化正规形所用的局部单测量恒等式。
verify_hunter_bound1_sharpness_n6.py	检查六符号八分量证书的覆盖、不交、简单性、强无提前离环性、权重、最小入权与 $B_1 = 863$ 。

文件	用途
<code>verify_preimage_chain_bound_n6.py</code>	重建 $F(P)$ ，在十九条四符号路径上检查精确恒等式，计算 Π ，并校准两个六符号例子。
<code>verify_preimage_component_matching.py</code>	独立重建 $F(P)$ 与 D_P ，穷尽三符号全部 720 条哈密顿路径，并以归档及确定性四符号样本审计源端、目标端对应。
<code>verify_full_primitive_block_stability.py</code>	审计局部门户状态、本原转移、精确整数层公式及直至十四阶的数值表。
<code>verify_revision_edge_cases.py</code>	检查全覆盖带符号偏差、 $k = 5$ 桥接歧义及 $k = 8, \dots, 14$ 的显式桥接构造。
<code>verify_self_nesting_bridge_macro_theory.py</code>	校准自嵌套、终端满片刚性、本原刚性与补充性的桥接宏实验。

规范有限审计命令为

```
python run_quick_audit.py
python run_full_audit.py
```

冻结参考环境为 Linux x86_64 上的 CPython 3.13.5，不使用第三方 Python 包。精确命令、主张-检查器映射、预期报告、发布元数据、最终 Lean 日志与 SHA-256 清单均包含在补充材料归档中。

层次	地位
Hunter–Raudvere 基线	固定外部 Lean 4.31.0 形式化
本文任意 k 的新主证明链	已合并发布的 Lean 4.31.0 形式化
六符号证书与 $k \leq 14$ 有限计算	确定性 Python 重建与审计
独立复现	不属于本文验证记录

13 结论与展望

前像链恒等式量化了 Hunter 分量被独立计价时损失的顺序信息。 σ^2 -约化正规形消除了零成本权二跃迁的歧义；费率缺陷正规形与端点匹配容量定理将有限缺陷预算转化为刚性终端门户族。固定实际像的点态基线为每条哈密顿路径赋予规范的非负整数层，容量一门户指派则在统一扣除一个源端/终端例外后完成全局计价。由此得到显式修正项 Γ_k 以及表 1 中的下界。

从实际分量几何到无条件全局 $S(k)$ 定理的主证明链已由 Lean 4.31.0 核验。形式化确认， G_s 中的单个减一项由两个互斥终端情形共享；当特选分量作为额外非根零成本目标时，其代价由精确根差 $\mu_{\max} - \mu_{C_0}$ 支付。

六符号情形区分了低价分量系统与低价哈密顿前像：显式 Hunter 像满足 $B_1 = 863$ ，但其任一前像在指派松弛下至少承担六个附加单位。该结论不推出 $S(6) \geq 875$ ，因为已知的 872 字母构造满足 $B_1 + \Pi = 866$ 。

桥接块构造与中性细分收缩定理界定了该方法的一类精确边界：单分量、单门户仿射包络不可能具有有界的缺陷系数，而沿既有权三接缝切分本身不会产生新的独立义务。该约束不适用于非线性、多状态、跨分量、前缀敏感或迭代变换方法。超越当前 $\Theta((k-4)!)$ 修正需要引入附加的高阶结构。

作者贡献

概念构建、方法设计、形式分析、软件、验证、研究实施、初稿撰写及修订：Xiaolong Liu。

利益冲突

作者声明不存在利益冲突。

数据与代码可用性

第 12 节标识的补充材料归档包含确定性验证器、紧凑证书、规范报告、精确命令、主张映射、最终 Lean 发布记录、选定证明源码、构建与公理审计日志以及 SHA-256 清单。完整形式化工程由 [公开仓库](#) 及源码快照 `a3852f0de2d84b99cc65d4c281e86892b5f6e911` 精确标识；仓库同时提供中英文论文与补充材料。

生成式人工智能与人工智能辅助技术使用声明

本研究在准备过程中使用 ChatGPT Pro (OpenAI) 辅助探索性推导、计算原型、Lean 与 Python 代码开发以及语言编辑。全部数学主张、证明、引文、源码与结论均由 Xiaolong Liu 审核，并由其对发表内容承担完整责任。第 12 节所述 Lean 形式化与确定性计算审计构成本文的验证记录，但不等同于独立同行评审。

A 门户引理的坐标化证明

令 $n = k - 1$ 。一个片段的起始词记为 $z = z_0 \cdots z_{k-1}$ ；其标记字母为 z_{k-1} ，循环基词为 $z_0 \cdots z_{k-2}$ 。大小为 s 的片段包含

$$\mathcal{P}(z, s) = \{[z_{k-1}\sigma^i(z_0 \cdots z_{k-2})] : 0 \leq i < s\}. \quad (49)$$

其端点为 $z_{k-1}\sigma^{s-1}(z_0 \cdots z_{k-2})$ 。一个精确权三后继保留端点去掉前三个字母后的后缀，并在末尾接上前三个字母的六种排列之一。

引理 A.1 (标记旋转类对齐). 设 W, W' 是同一字母表上的排列词, z 为指定字母. 每个旋转类中恰有一个以 z 开头的代表. 因此, $[W] = [W']$ 当且仅当把二者循环旋转到 z 位于首位后, 所得词逐字相同.

证明. 字母 z 在每个排列词中恰出现一次, 因而恰有一个循环旋转以 z 开头. 旋转类相等遂等价于这两个唯一代表相等. $\square$

A.1 满片段游程

将满片段的起始词写成 Yam , 其中 $|Y| = k - 2$. 直接代入(49)可见, 唯一支撑不交的满片段后继为

$$Q(Yam) = \sigma(Y)am. \quad (50)$$

$z, Qz, \dots, Q^{k-3}z$ 的支撑两两不交. 若 $Q^i z$ 与 $Q^j z$ 中的旋转类相等, 则对齐唯一字母 a, m 后, 强迫 $i = j \pmod{k-2}$. 下一个支撑回到第一个. 因此引理 7.1(i) 成立.

A.2 非满片段端点的六种后继方向

把亏缺为 δ 的非满片段规范化, 使其端点为 $mabR$, 其中 $|R| = k - 3$. 在标记字母为 m 、循环基词为 abR 的带标记的完全二循环中, 以 $\mathbb{Z}_n$ 标记关联位置, 并令端点旋转类的坐标为 0. 该片段占据

$$I_\delta = \{0, \delta + 1, \delta + 2, \dots, n - 1\}; \quad (51)$$

遗漏循环弧为 $J_\delta = \{1, \dots, \delta\}$. 六个精确权三后继起始词为 $R\pi(m, a, b)$. 若 $\pi(m, a, b) = (p, q, t)$, 则经过 j 次满片段映射 Q 后, 候选满片段的起始词为

$$\sigma^j(Rp)qt,$$

支撑为

$$\mathcal{F}_j(p, q, t) = \{[t\sigma^\ell(\sigma^j(Rp)q)] : \ell \in \mathbb{Z}_n\}. \quad (52)$$

由引理 A.1, 判断两个旋转类是否相同, 可以先把二者旋转到同一指定字母位于首位. 由于 R 中各字母的循环次序保持不变, 一旦 m, a, b 的位置确定, 其余字母便强迫唯一的一对关联下标 (i, ℓ) . 解出六组初等词等式, 得到下表. j 为候选满片段序号, 从 $j = 0$ 开始; 所有下标均按模 n 理解.

(p, q, t)	$j_{\min}$	一组强迫等式 (i, ℓ)	可连续出现的不交满片段数
(m, a, b)	0	$(0, 0)$	0
(m, b, a)	0	$(0, n - 1)$	0
(a, b, m)	0	$(\delta + 1, \delta - 1)$	0
(a, m, b)	$\delta - 1$	$(\delta + 1, n - \delta)$	$\delta - 1$
(b, m, a)	$\delta - 1$	$(\delta + 1, n - \delta - 1)$	$\delta - 1$
(b, a, m)	$n - 2$	$(\delta + 1, \delta)$	$n - 2 = k - 3$.

记号 (i, ℓ) 表示

$$[m\sigma^i(abR)] = [t\sigma^\ell(\sigma^{j_{\min}}(Rp)q)].$$

前三种方向在 $j = 0$ 即发生所列交叠。对 (a, m, b) 与 (b, m, a) ，对齐有序子词 R 强迫 $j \geq \delta - 1$ ，表中下标给出 $j = \delta - 1$ 时的等式，故此前不存在其他交叠。对 (b, a, m) ，同一对齐强迫 $j \equiv n - 2 \pmod{n}$ ，所以 $j = 0, \dots, n - 3$ 均不交，而 $j = n - 2$ 首次返回。上述必要性排除了所有其他旋转等价，所列等式则证明充分性。这证明引理 7.1(iii)。

当 $\delta = 1$ 时，第四、第五行长度为零。唯一正长度满片段游程使用最后一种方向，其边界满片段包含唯一遗漏的关联位置；同一亏缺为一的非满片段另一侧若存在正长度满片段游程，也必须使用该关联位置，因而两个边界满片段相交。这证明引理 7.1(ii)。

A.3 内部长游程阈值

设左非满片段亏缺为 a 。沿唯一长方向前进 $n - 2 = k - 3$ 个满片段，并把所得头顶点规范化为 $macU$ 。右非满片段的六个可能的起始词为 $U\pi(m, a, c)$ 。其中五种方向都有显式公共类：

$\pi(m, a, c)$	已经属于累计支撑的显式旋转类
(m, a, c)	$[macU]$
(m, c, a)	$[macU]$
(a, m, c)	$[mcUa]$
(a, c, m)	$[mUac]$
(c, m, a)	$[maUc]$.

剩余方向为 (c, a, m) ，其标记字母为 m 。把左右两个边界片段运输到同一带标记关联循环，并选择坐标使左占据区间为

$$L_a = \{0, a + 1, a + 2, \dots, n - 1\},$$

右占据区间为

$$R_b = \{1, 2, \dots, n - b\}.$$

于是

$$L_a \cap R_b = \{a + 1, a + 2, \dots, n - b\}, \quad |L_a \cap R_b| = \max(0, n - a - b).$$

两个边界片段不交当且仅当

$$a + b \geq n = k - 1. \quad (53)$$

前五种方向已有上表中的显式公共类。因此， (c, a, m) 是唯一可能不交的方向；这也证明了引理 7.1(iv) 中阈值的必要性与充分性。

B 显式双桥接块构造

以下设 $k \geq 8$, 令 $n = k - 1$, 标记基词为 $B = (b_0, \dots, b_{n-1})$, 标记字母为 m 。从 Bm 出发, 先取一个大小为 $k - 3 = n - 2$ 的非满片段。第一个满片段的起始词显式规定为

$$z_1 = (b_{n-1}, b_0, \dots, b_{n-4}, b_{n-2}, b_{n-3}, m). \quad (54)$$

直接代入重叠定义可知, 从首个非满片段的头到 z_1 的接缝权恰为三, 且二者支撑不交。随后再取 $k - 5 = n - 4$ 个满片段; 从这一刻起, 每一步才使用(50)中已经证明的“满片段到满片段”唯一 Q -后继。所得片段大小为

$$(k - 3), (k - 1)^{k-4},$$

并定义桥接块 $B_k(B, m)$ 。该构造不要求从首个非满片段到首个满片段的后继具有唯一性。

端点计算给出一个具有相同标记字母的精确权三后继, 其基词为

$$\Theta(B) = (b_{n-4}, b_{n-2}, b_{n-1}, b_0, \dots, b_{n-5}, b_{n-3}), \quad (55)$$

支撑不交性由下述论证给出。由于标记字母 m 唯一, 两个以 m 开头表示的旋转类相等, 当且仅当其长度为 n 的基词完全相等。把锚字母 $a = b_{n-3}$ 从源桥接的各循环中删去。每个源片段都由固定循环骨架

$$L = (b_0, b_1, \dots, b_{n-4}, b_{n-2}, b_{n-1})$$

在某个循环间隙中插入 a 得到, 唯一未使用的循环间隙为

$$(b_{n-5}, b_{n-4}). \quad (56)$$

确切地说, 第一个非满片段在 b_{n-4} 与 b_{n-2} 之间插入 a ; 后续满片段依次在

$$(b_{n-2}, b_{n-1}), (b_{n-1}, b_0), (b_0, b_1), \dots, (b_{n-6}, b_{n-5})$$

之间插入它。对目标基词 $B' = \Theta(B)$, 其可变锚字母为 $a' = b_{n-6}$ 。删去 a' , 再把剩余循环骨架旋转到从 b_0 开始, 得到

$$L' = (b_0, \dots, b_{n-7}, b_{n-5}, a, b_{n-4}, b_{n-2}, b_{n-1}).$$

进一步同时删去 a, a' 后, 源、目标拥有同一个公共循环骨架。在每个目标桥接循环中, 字母 a 固定在循环间隙 (b_{n-5}, b_{n-4}) 中, 因为目标各片段只改变 a' 的插入位置。故一个目标循环若等于源循环, 源必须恰在(56)中插入 a ; 但这正是源桥接块唯一缺失的循环间隙。因此两个桥接支撑不交。

端点前三个字母的另外五种排列已经与源最后满片段支撑共享旋转类, 如命题 11.1 的六方向比较所示。因此(55)给出所需的不交后继。两个桥接块合并后有 $r = 2(k - 3)$ 、 $\Delta = 4$ 、 $x = 0$ 、 $\mu = 2$ 。由(17)与 $t = (k - 1)r - \Delta$ 得到命题 11.2 中的参数。

C 分层修正项的精确最小化

令

$$M = (k-1)!, \quad D = D_k, \quad E = (k-1)(k-3).$$

令 Z_0 为(38)在 $s = 0$ 时的值。由于 $D + 1 = (k-1)(k-2)$ 且 $C_s = C_0 + (D+1)Ds$,

$$Z_s = \max(0, Z_0 - (k-1)s). \quad (57)$$

又因为 $D = E + (k-2)$,

$$\left\lfloor \frac{\kappa_k + sD}{E} \right\rfloor = s + \left\lfloor \frac{\kappa_k + s(k-2)}{E} \right\rfloor. \quad (58)$$

所以

$$G_s = \max \left(0, Z_0 - 1 - ks - \left\lfloor \frac{\kappa_k + s(k-2)}{E} \right\rfloor \right). \quad (59)$$

命题 C.1 (精确最优层). Γ_k 等于满足

$$ks + \left\lfloor \frac{\kappa_k + s(k-2)}{E} \right\rfloor \geq Z_0 - 1 \quad (60)$$

的最小整数 $s \geq 0$ 。

证明. 当 $G_s > 0$ 时, (43)与(59)给出

$$s + G_s = Z_0 - 1 - (k-1)s - \left\lfloor \frac{\kappa_k + s(k-2)}{E} \right\rfloor,$$

它随 s 严格下降。一旦 $G_s = 0$, 目标函数等于 s , 随 s 严格上升。在前一层, (60)作为整数不等式至少差一, 因此其目标值不小于第一个零缺口层。□

最后,

$$Z_0 = \frac{M}{D} + O(k), \quad (61)$$

因为 $0 \leq \kappa_k < D$ 。由(60),

$$\frac{Z_0 - 1 - \kappa_k/E}{k + (k-2)/E} \leq \Gamma_k \leq \left\lceil \frac{Z_0 - 1}{k} \right\rceil.$$

因此

$$\Gamma_k = \frac{(k-1)!}{kD_k} + O(1) = \left(1 - \frac{3}{k} + O(k^{-2}) \right) (k-4)!, \quad (62)$$

因为

$$\frac{(k-1)!}{kD_k(k-4)!} = \frac{(k-1)(k-2)(k-3)}{k(k^2 - 3k + 1)} = 1 - \frac{3}{k} + O(k^{-2}).$$

参考文献

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